Noise Covariance Defines Reconstruction Geometry: Optimal Filtering, Weighted Subspaces, and Tied Linear Autoencoders

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Abstract

Reconstructing detector signals under colored or correlated noise requires a geometry consistent with the measurement likelihood. Under a linear Gaussian model, the inverse covariance Σ^{-1} defines that geometry: optimal filtering (OF) is a fixed rank-1 projection, expectation–maximization PCA (EMPCA) learns a rank-k projection, and a tied noise-aware linear autoencoder recovers the same weighted subspace. Isotropic mean-squared error therefore represents the likelihood only for white noise. Nine controlled tests supported the predicted normalization, equivalence, covariance, and residual-calibration behavior; eight used ten seeds and the colored-noise metric-reversal test used a confirmatory 30-seed held-out analysis. On seven public CRESST-A channels, pulse and noise streams were each split chronologically, separately. Rank-6 EMPCA reduced the held-out weighted residual relative to single-template OF by 1012.27 over 19,335 pulses (pooled event-bootstrap 95% CI $[927.90, 1097.61]$) and met the predeclared multi-template-OF competitiveness gate. A post-freeze channel-then-event analysis retained positive equal- channel OF and same-rank PCA contrasts, with hierarchical 95% intervals $[110.69, 1537.48]$ and $[1.22, 6.69]$; only seven clusters limit broader inference. The result supports within-channel linear reconstruction, not cross-channel transfer or detector sensitivity. In a public GWOSC boundary test, PSD and shared-FIR implementation checks passed, but held-out calibration failed for both global and predeclared 64-second local PSD models on one GW150914-centered interval. Event and injection

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scores are therefore diagnostic only. Because neither release records the true signal, we add a second public real-detector dataset, TIDMAD/ABRACADABRA, whose validation files carry an injected reference channel: all six estimator-identity gates passed on real SQUID noise, and a simple global Gaussian PSD model calibrated it (χ^2/dof 1.003, held-out score dispersion 1.014) where the same class of model failed on strain. On the same release the rank-2 quadrature prediction was not confirmed and the linear estimator scored below the no-denoising baseline on the release's own detector-facing benchmark. The evidence separates the likelihood geometry implied by a covariance model from the empirical adequacy of that model on real noise.

1 Introduction

Scientific detectors often record a structured response superposed with electronic, thermal, or environmental fluctuations. These fluctuations are rarely independent and identically distributed. They may be colored in frequency, heteroskedastic across samples, correlated between readout channels, or slowly varying in time. Reconstruction must therefore do more than smooth a waveform: it must estimate signal amplitudes or shapes without spending model capacity on directions dominated by detector noise.

Three communities describe closely related solutions in different language. Signal processing uses matched or optimal filters to project a trace onto a known template with inverse-noise weighting [1, 2]. In cryogenic calorimetry, this construction is the standard optimal-filter (OF) amplitude estimator [3–5]. Statistics replaces the fixed template by a learned low-dimensional signal model. Probabilistic PCA assumes isotropic noise [6], factor analysis allows diagonal noise [7], and weighted or expectation–maximization PCA (EMPCA) accommodates nonuniform weights [8, 9]. Machine learning expresses a similar projection as an encoder, a latent representation, and a decoder; under Euclidean mean-squared error, tied linear autoencoders recover the PCA subspace [10–12].

The common object is the observation-space metric. For a trace $x = s + n$ with $n \sim \mathcal{N}(0, \Sigma)$, the negative log-likelihood of a candidate reconstruction $\hat{s}$ contains $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$. Thus Σ^{-1} determines which residual directions are costly. OF, EMPCA, and the tied noise-aware linear autoencoder can all be viewed as projections in this metric. They differ in their admissible signal model: a known one-dimensional template for OF and a learned k-dimensional subspace for EMPCA and the tied linear autoencoder. This viewpoint separates three choices that are often conflated: the likelihood determines residual geometry, the signal model determines admissible reconstructions, and the algorithm or architecture parameterizes the projection.

This distinction matters because isotropic mean-squared error (MSE) is not a neutral default. It is the Gaussian likelihood only when $\Sigma = \sigma^2 I$. Under colored or correlated

noise, MSE ranks residuals in a different geometry and can favor variance-rich but low-information directions. A method may therefore reduce raw MSE while worsening the covariance-weighted residual that follows from the assumed detector model. The effect is a likelihood misspecification, not merely a change of units.

Our theoretical contribution is a compact hierarchy of constrained Gaussian likelihood projections. We identify classical OF as the fixed rank-1 case, state the exact finite-sample degeneracy of a rank-1 weighted subspace projection to OF when its basis spans the template, and give the corresponding population result for a learned rank-1 EMPCA direction. We then establish a bridge from weighted subspace estimation to representation learning: a tied linear autoencoder whose encoder is the Σ^{-1}-adjoint of its decoder and whose loss is covariance-weighted has the same optimal subspace as EMPCA. Full derivations, algorithmic updates, conditioning details, and gauge arguments are placed in the supplement so that the main text remains focused on the likelihood geometry.

The empirical contribution follows a deliberately asymmetric evidence arc. Controlled simulations first test consequences that are known under the data- generating model. Public CRESST-II/III pulse-shape data [13] then provide a positive but narrow real-detector test: within each of seven channels, we compare fixed-template and learned linear reconstructions on held-out pulses. Pulse and noise streams were each split chronologically, separately; no common cross-stream timestamp cutoff is implied. A post-freeze channel-then-event bootstrap tests whether the pooled result survives equal channel weighting. The OF–EMPCA contrast tests the benefit of enlarging the admissible signal model, whereas the same-rank PCA–EMPCA contrast and covariance controls test the contribution of the noise geometry. We keep these questions separate because a gain over a fixed template is not, by itself, evidence that inverse-covariance weighting caused the gain. Public H1/L1 data around GW150914 from the Gravitational Wave Open Science Center (GWOSC) [14] then provide a negative calibration stress test. Neither of these releases records the signal that was actually present, so both compare residuals rather than distance to truth. We therefore add public TIDMAD data from the ABRACADABRA axion search [15], whose validation files carry an injected reference channel alongside the detector readout: it supplies the real-data ground truth the other two lack, and its noise model calibrates where GWOSC's does not. This order distinguishes reconstruction performance from detection sensitivity and implementation identity from statistical calibration.

The scope is linear and Gaussian. The exact statements do not extend to nonlinear autoencoders, non-Gaussian transients, or a covariance model that is incorrect for the evaluation data. CRESST is used only to assess within-channel reconstruction of public held-out pulses, not cross-channel transfer or detector sensitivity. GWOSC is used to test whether specific global and local PSD models calibrate a matched statistic on one public interval, not to estimate GW150914 significance or general sensitivity. TIDMAD

is used at one frozen operating point on named validation files; it is not a survey of the release, it does not confirm the rank prediction on real data, and the linear estimator is not competitive on the release's own detector-facing denoising score.

The remainder of the paper introduces the common objective and whitening map, treats OF and EMPCA as rank-constrained cases, states the tied-linear- autoencoder bridge and the consequence for MSE, and then presents synthetic, CRESST, GWOSC, and TIDMAD evidence in that order. The discussion separates supported conclusions from limitations. Supplementary material contains the extended OF derivations, full proofs, EM implementation details, symmetry and gauge analysis, evidence provenance, and the status of deferred supplementary diagnostics.

2 The Unified Noise-Aware Objective

This section formulates the reconstruction problem as a single Gaussian maximum-likelihood (ML) objective and shows that optimal filtering (OF) and expectation–maximization PCA (EMPCA) are constrained solutions of it. We distinguish three levels of equivalence used throughout the paper: *estimator equivalence* (identical scalar amplitude estimates under matched preprocessing and weighting), *subspace equivalence* (identical signal subspace up to gauge freedom), and *amplitude equivalence* (identical coefficients once projected onto the same subspace with consistent normalization).

Notation and conventions. Vectors are columns. We work in $\mathbb{C}^d$ to cover frequency-domain representations; $\dagger$ denotes the conjugate transpose, and all statements specialize to real-valued data by replacing $\dagger$ with the transpose T. Throughout, $x \in \mathbb{C}^d$ is one observed trace, s its unknown noiseless signal, $\hat{s}$ a candidate reconstruction, s_0 a fixed signal template, $\Sigma \succ 0$ the noise covariance with inverse Σ^{-1} and whitening map $\Sigma^{-1/2}$, $U \in \mathbb{C}^{d\times k}$ a learned rank-k basis with coefficient vector $c \in \mathbb{C}^k$, and $P \in \mathbb{C}^{d\times k}$ the decoder basis of the linear autoencoder. Whitened quantities carry tildes, e.g. $\tilde{x} = \Sigma^{-1/2}x$. Events are indexed by $i = 1, \ldots, N$, frequency bins by k (the rank is also written k; the meaning is always clear from context), and channels by $c = 1, \ldots, C$ where needed.

2.1 Generative model

For each event i we observe $x_i \in \mathbb{C}^d$, a time-domain trace or its frequency-domain coefficients, with the generative model

$$x_i = s_i + n_i, \qquad n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma), \qquad \Sigma \succ 0, \tag{1}$$

where s_i is the unknown noiseless signal for event i and the covariance Σ is known or estimated from dedicated noise data that are independent of the fitted trace. The reconstruction $\hat{s}_i$ is restricted to a specified signal model. In the fixed-template special case, $s_i = a_i s_0$ and $\hat{s}_i = \hat{a}_i s_0$ for a scalar amplitude a_i.

Define the whitening transform

$$\tilde{x}_i = \Sigma^{-1/2} x_i, \qquad \tilde{s} = \Sigma^{-1/2} s_0. \tag{2}$$

In whitened coordinates the noise is isotropic,

$$\tilde{x}_i = \Sigma^{-1/2} s_i + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}_{\mathbb{C}}(0, I), \tag{3}$$

so every Σ^{-1}-weighted projection becomes an ordinary orthogonal projection. Stacking events row-wise gives the whitened data matrix $\tilde{X} \in \mathbb{C}^{N \times d}$ with rows $\tilde{x}_i^\dagger$.

2.2 The Gaussian ML objective

For a single trace x under model (1), the Gaussian negative log-likelihood is, up to additive constants,

$$\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s}). \tag{4}$$

Every estimator in this paper minimizes objective (4); the methods differ only in the constraint placed on the admissible reconstruction $\hat{s}$.

Optimal filtering: fixed rank-1 constraint. For a known template s_0 and unknown scalar amplitude A, the constraint is $\hat{s} = A s_0$. Minimizing (4) over A gives the generalized least-squares estimator

$$\hat{A}_{\mathrm{OF}} = \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0}, \tag{5}$$

which is exactly the optimal filter in covariance form. For stationary noise in Fourier space, Σ is diagonal and (5) reduces to the familiar PSD-weighted OF formula (Section 3).

EMPCA: learned rank-k constraint. For a learned basis $U = [u_1, \ldots, u_k] \in \mathbb{C}^{d \times k}$, the constraint is $\hat{s} = Uc$, and

$$\chi^2(c) = (x - Uc)^\dagger \Sigma^{-1} (x - Uc) \tag{6}$$

is minimized by the weighted normal equations

$$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x, \qquad \hat{s}_{\mathrm{EMPCA}} = U\hat{c}. \tag{7}$$

Equation (7) makes the connection to OF explicit: OF is the special case $k = 1$ with $U = s_0$ fixed.

2.3 Whitening as the geometric normal form

Whitening converts the Σ^{-1}-weighted inner product into the standard Euclidean one: for any $u, v \in \mathbb{C}^d$,

$$u^\dagger \Sigma^{-1} v = (\Sigma^{-1/2} u)^\dagger (\Sigma^{-1/2} v) = \tilde{u}^\dagger \tilde{v}. \tag{8}$$

After whitening, the noise covariance is the identity and all estimators in this paper become ordinary orthogonal projections; this is the simplification that makes the geometric statements of Sections 3–5 exact.

In the most common detector setting the noise is wide-sense stationary, so Σ is circulant and diagonalized by the DFT:

$$\Sigma = F^\dagger \mathrm{diag}(J_1, \ldots, J_d) F, \tag{9}$$

where $J_k = \mathbb{E}[|N_k|^2]$ is the noise power spectral density (PSD) at frequency bin k. Whitening is then element-wise division in the frequency domain,

$$\tilde{X}_k = X_k / \sqrt{J_k}. \tag{10}$$

For non-circulant Σ (short traces, non-stationary noise, correlated channels), $\Sigma^{-1/2}$ is computed by eigendecomposition or Cholesky factorization; numerically small PSD bins require a regularization floor or frequency masking. These implementation details, which do not affect the theory, are collected in Appendix C.

2.4 The estimator hierarchy

Table 1 summarizes the estimator family studied in this paper. Each row is a constrained minimizer of the same Gaussian ML objective (4); the rows differ only in how the admissible reconstruction is constrained.

Table 1: Hierarchy of constrained ML estimators under the unified noise-aware objective. All methods minimize $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$. "Degeneracy condition" states when a row coincides with the basic OF row.

Estimator	Constraint on $\hat{s}$	Subspace dim.	Solution form	Degeneracy condition
Basic OF	$\hat{s} = A s_0,\ s_0$ fixed	$k = 1$ (fixed)	$\hat{A} = (s_0^\dagger \Sigma^{-1} x)/(s_0^\dagger \Sigma^{-1} s_0)$	Signal is exactly $\propto s_0$
Time-shift OF	$\hat{s} = A e^{-i\omega t_0} s_0,\ t_0$ unknown	$k = 1$ (shift manifold)	$\hat{t}_0 = \arg\max_{t_0} \lvert\hat{A}(t_0)\rvert$	No timing jitter; collapses to basic OF
Joint-channel OF	$\hat{s} = A s,\ s \in \mathbb{C}^{Cd}$ joint template	$k = 1$ (joint)	$\hat{A} = (s^\dagger \Sigma^{-1} x)/(s^\dagger \Sigma^{-1} s)$	Channels uncorrelated; factorizes per channel
Rank-k EMPCA	$\hat{s} = U c,\ U \in \mathbb{C}^{d \times k}$ learned	k (learned)	$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x$	$k = 1$, $\mathrm{span}(U) = \mathrm{span}(s_0)$: equals basic OF
Noise-aware linear AE	$\hat{s} = P P^\dagger \Sigma^{-1} x,\ P^\dagger \Sigma^{-1} P = I$	k (learned)	$P^\star$ = top-k eigenvectors of $\hat{R}_x \Sigma^{-1}$	Equivalent to EMPCA at convergence (Theorem 5.1)

Table 2 states the same comparison from an evaluation perspective. Ordinary PCA is included because it is the Euclidean baseline used in the loss-geometry experiments, even

Table 2: Methods compared under a shared reconstruction task. "Uses covariance" refers to the fitted objective, not merely to post-hoc evaluation.

Method	**Model and output**	**Uses Σ**	**Known template**	**Colored or correlated noise**	**Likelihood interpretation**
OF / matched filter	Fixed rank-1; amplitude and template reconstruction	Yes	Yes	Yes, with valid Σ or PSD	Gaussian ML for the template amplitude
Ordinary PCA	Learned rank-k Euclidean subspace and scores	No	No	Only after exact whitening	Gaussian ML only for $\Sigma = \sigma^2 I$
Weighted PCA / EMPCA	Learned rank-k weighted subspace and coefficients	Yes	No	Yes	Gaussian ML for the specified covariance or weights
Noise-aware tied linear AE	Rank-k subspace, latent code, and reconstruction	Yes	No	Yes, in this paper	Same linear-Gaussian objective as weighted PCA / EMPCA

though it does not minimize the covariance-weighted objective unless the data have first been correctly whitened.

Three structural observations follow. First, constraint complexity increases monotonically down the table: basic OF is the most constrained (one free parameter, fixed geometry) and the noise-aware linear AE the least (a full learned k-dimensional subspace). Second, the noise model enters identically in every row through the same quadratic form — the methods differ in their signal model, not in how they treat noise. Third, degeneracy propagates upward: each row collapses to the row above when its extra freedom is unused. Sections 3–5 prove these degeneracies; exact improvement over OF occurs only when the true signal family has dimensions a fixed template cannot capture.

2.5 Multi-channel extension

Detectors with C simultaneously read-out channels are covered by stacking the channel traces into a joint vector $x \in \mathbb{C}^{Cd}$ with block-structured covariance $\Sigma_{\text{joint}} \in \mathbb{C}^{Cd \times Cd}$, whose diagonal blocks are intra-channel covariances and whose off-diagonal blocks carry cross-channel noise correlations. The Gaussian ML objective (4) is unchanged in form; the rank-1 constraint gives joint-channel OF (Section 3.3) and the rank-k constraint gives multi-channel EMPCA. When cross-channel correlations vanish and each channel covariance is

diagonal, the joint objective decouples into C independent single-channel problems. The full block-covariance construction and whitening details are given in Appendix B.2.

3 Optimal Filtering as Rank-1 Constrained ML

3.1 The rank-1 constraint and OF estimator

Specializing the unified objective (4) to the fixed-template constraint $\hat{s} = As_0$ in the stationary single-channel case, where Σ is diagonal in frequency space with PSD entries J_k, gives

$$\chi^2(A) = \sum_k \frac{|X_k - AS_k|^2}{J_k}, \tag{11}$$

the discrete Gaussian log-likelihood for uncorrelated frequency bins [2]; this frequency-domain form for microcalorimeters was first stated by Szymkowiak et al. [3]. The stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A} = \frac{\displaystyle\sum_k \frac{S_k^* X_k}{J_k}}{\displaystyle\sum_k \frac{|S_k|^2}{J_k}}, \tag{12}$$

which is the covariance-form estimator (5) in frequency coordinates. Defining the filter kernel and normalization

$$\Phi_k = \frac{S_k^*}{J_k}, \qquad N_\Phi = \sum_k \frac{|S_k|^2}{J_k}, \qquad \Phi_k^{(0)} = \frac{\Phi_k}{N_\Phi}, \tag{13}$$

the estimate $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ is the filtered inner product of the data with the normalized kernel — the discrete matched filter [1]. Geometrically, $\hat{s}_{\mathrm{OF}} = \hat{A}s_0$ is the Σ^{-1}-orthogonal projection of x onto the fixed one-dimensional template subspace.

Assumption boundary. The covariance-form estimator requires a known template direction and a known or independently estimated positive-definite noise covariance. Stationarity is needed only for the convenient diagonal PSD representation in Eq. (11); the generalized least-squares expression remains valid for non-stationary or cross-channel covariance when the full matrix is used. Gaussianity is required for the likelihood and efficiency claims, while the best-linear-unbiased statement needs only the corresponding second-moment assumptions. OF is therefore optimal for this fixed-template linear problem. It is not a solution to general nonlinear waveform reconstruction, unknown signal morphology, or covariance mismatch.

3.2 Estimator variance and energy resolution

The OF estimator is unbiased, and its variance

$$\mathrm{Var}(\hat{A}) = \frac{1}{N_\Phi}, \qquad \sigma_A = \frac{1}{\sqrt{N_\Phi}}, \tag{14}$$

attains the Cramér–Rao bound (CRB) for the Gaussian model, making OF the minimum-variance unbiased estimator for a fixed template; the full Fisher information calculation is given in Appendix A. When the amplitude calibrates to deposited energy, $A = E/E_0$, the baseline energy resolution is

$$\sigma_E = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}. \tag{15}$$

Equation (15) carries the design intuition: resolution improves where signal power is large and noise PSD small, and N_Φ is simultaneously the filter normalization and the total signal-to-noise budget available to any rank-1 estimator. Empirical verification of the CRB and of the $\sigma_E \propto \sqrt{\text{noise power}}$ scaling is reported in Appendix F.

3.3 Extensions: the same projection principle

Two standard OF extensions are structured applications of the same rank-1 weighted projection principle, and both are absorbed naturally by the hierarchy of Table 1.

Time-shift OF. For an unknown arrival time, the admissible signals form the one-parameter manifold $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$. Profiling the amplitude at each candidate shift gives $\hat{A}(t_0) = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k$ — the inverse DFT of the filtered spectrum — and the ML delay maximizes the filtered amplitude, $\hat{t}_0 = \arg\max_{t_0} |\hat{A}(t_0)|$ [5]. For small shifts, a Taylor expansion shows $\mathcal{M}$ is approximately contained in $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$, so a low-rank learned subspace can absorb timing jitter; this is exactly the mechanism by which rank-2 EMPCA removes timing bias in Section 7. The full profiling derivation and the formal degeneracy statement (Proposition B.1) are given in Appendix B.1.

Joint-channel correlated OF. For C channels stacked into $x, s \in \mathbb{C}^{Cd}$ with full joint covariance Σ, the GLS estimator $\hat{A}_{\mathrm{JOF}} = (s^\dagger \Sigma^{-1} x)/(s^\dagger \Sigma^{-1} s)$ is algebraically the rank-1 weighted subspace estimator (7) with $U = s$. Cross-channel noise correlations are handled identically in OF and EMPCA, through the same quadratic form; the formal statement (Proposition B.2) and the block-covariance details are in Appendix B.2.

3.4 Limitation: the fixed-template assumption

The OF derivation assumes every signal is a scalar multiple of the *same* known template. Realistic pulse families violate this in two generic ways: timing jitter spreads the signal over the shift manifold, and physical effects (position-dependent response, geometry inhomogeneity, channel cross-talk, saturation) make the pulse shape itself vary event-to-event. When the template direction s_0 is misaligned with the true signal s, the OF estimate is biased,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} \neq A \quad (\text{if } s_0 \not\propto s), \tag{16}$$

with effective SNR loss $\cos^2\theta_w$, the squared cosine of the whitened-space angle between s_0 and s. A template-mismatch experiment quantifying this bias is reported in Appendix F. Replacing the fixed template by a *learned* k-dimensional subspace removes this assumption: when the true signal family lies in the learned subspace, the estimator is unbiased and attains the CRB of the subspace-constrained problem. This motivates the rank-k extension developed next.

4 EMPCA as Rank-k Constrained ML

4.1 The rank-k constraint

The optimal filter assumes a one-dimensional signal family. Realistic families have more structure: pulse shapes vary with interaction position, timing jitter generates a continuum of shifted templates, and multi-mode excitations populate several directions simultaneously. The natural relaxation constrains the signal to a *learned* k-dimensional subspace. For a single observation $x \in \mathbb{C}^d$ the rank-k constrained ML problem is

$$\min_{U,\,c} \ \chi^2(c;\,U) = (x - Uc)^\dagger \Sigma^{-1} (x - Uc), \qquad \text{subject to} \quad U^\dagger \Sigma^{-1} U = I_k, \tag{17}$$

and over N events with coefficient matrix $C = [c_1, \ldots, c_N] \in \mathbb{C}^{k \times N}$,

$$\min_{U,\,C} \ \mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} (x_i - Uc_i)^\dagger \Sigma^{-1} (x_i - Uc_i), \qquad U^\dagger \Sigma^{-1} U = I_k. \tag{18}$$

In whitened coordinates ($\tilde{x}_i = \Sigma^{-1/2} x_i$, $\tilde{U} = \Sigma^{-1/2} U$, $\tilde{U}^\dagger \tilde{U} = I_k$) the objective reduces to a standard low-rank approximation problem,

$$\mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} \|\tilde{x}_i - \tilde{U} c_i\|_2^2, \tag{19}$$

whose minimizer over subspaces is the span of the top-k eigenvectors of the whitened sample covariance $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, with $\hat{R}_x = N^{-1}\sum_i x_i x_i^\dagger$. Standard PCA is the ML estimator of this subspace only when $\Sigma = \sigma^2 I$; EMPCA restores ML optimality for general Σ by carrying the weight Σ^{-1} through the estimation.

Algorithm. EMPCA solves (18) by alternating minimization [8]: the E-step solves the weighted normal equations for each c_i with U fixed (closed form, Eq. (7)), and the M-step solves a weighted regression for U with C fixed, followed by re-orthonormalization in the Σ^{-1} inner product. The objective (18) is biconvex, each step is an exact block minimization, and the iteration decreases the objective monotonically and converges to the stationary set, which for this problem coincides with the top-k eigenspace up to gauge (Theorem C.1, Appendix C.5). Implementation details — the frequency-domain variant of Yu et al. [9], initialization, model-order selection, and conditioning safeguards — are collected in Appendix C.

Ordinary PCA versus weighted PCA / EMPCA. Ordinary PCA minimizes Euclidean reconstruction error in the original coordinates. The weighted problem above minimizes that error after the noise geometry has been applied. When every event shares the same positive-definite covariance Σ, exact whitening reduces the weighted problem to ordinary PCA on $\tilde{X} = X\Sigma^{-1/2}$; this is an equivalence between weighted PCA in detector coordinates and ordinary PCA in whitened coordinates. It is not, in general, an equivalence to ordinary PCA on the unwhitened traces. The latter coincides with the weighted solution when $\Sigma = \sigma^2 I$, or accidentally when the weighting leaves the relevant eigenspace and its ordering unchanged. With event-dependent variances, missing samples, or weights Σ_i^{-1}, no single global whitening need exist and the alternating EMPCA updates are required. Consequently, PCA and EMPCA should agree in the white-noise control but should not be expected to agree under genuinely colored weighting. Numerically identical outputs in that regime would require an explicit audit of whether the weights entered the fit, whether both methods received already whitened data, and whether a degenerate eigenspace obscured the comparison.

Gauge freedom. The basis U is identified only up to a unitary rotation $U \mapsto UQ$, $Q \in \mathcal{U}(k)$: the reconstruction $\hat{s} = Uc$ and the subspace are gauge-invariant, the individual coefficients are not. Physical amplitude estimates therefore use gauge-invariant functionals, either the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}}$ or the projection onto a calibration template,

$$\hat{A} = \frac{s_0^\dagger \Sigma^{-1} \hat{s}}{s_0^\dagger \Sigma^{-1} s_0}, \tag{20}$$

which reduces to the OF estimator when $k = 1$ and $\mathrm{span}(U) = \mathrm{span}(s_0)$. The geometric structure behind this gauge freedom (the Grassmannian picture of subspace optimization) is developed in Appendix E.

4.2 Equivalence theorems: rank-1 EMPCA and basic OF

We can now state the first main result. Recall the three levels of equivalence from Section 2: subspace equivalence implies amplitude equivalence implies estimator equivalence, provided weighting and normalization conventions are matched.

Theorem 4.1 (Rank-1 OF–EMPCA equivalence, population level)**.** *Assume (i) the generative model $x_i = a_i s + n_i$ with fixed template s; (ii) Gaussian noise $n_i \sim \mathcal{N}(0, \Sigma)$ with known covariance; (iii) identical whitening, baseline subtraction, and preprocessing conventions; and (iv) a one-dimensional true signal subspace. Then, at the population level (sample covariance equal to its expectation):*

1. *the rank-1 EMPCA subspace equals* $\mathrm{span}(\tilde{s})$ *in whitened space;*
2. *the EMPCA coefficient estimator equals the generalized least-squares / optimal-filter estimator;*
3. *hence rank-1 EMPCA and OF are estimator-equivalent.*

Proof sketch. In whitened space the population covariance is the spiked identity $\mathbb{E}[\tilde{x}\tilde{x}^\dagger] = \mathbb{E}[a^2]\,\tilde{s}\tilde{s}^\dagger + I$, whose leading eigenvector is $\tilde{s}/\|\tilde{s}\|$. Rank-1 EMPCA recovers this eigenvector, and the rescaled projection coefficient is $\hat{a}_i = (s^\dagger \Sigma^{-1} x_i)/(s^\dagger \Sigma^{-1} s)$, which is exactly (5). The full proof, including the eigenvalue gap and the normalization bookkeeping, is given in Appendix D.1. □

Theorem 4.1 is a population-level statement; finite-sample deviations are controlled by the Davis–Kahan bound discussed in Section 4.4. The next result is exact at any sample size, for a fixed basis.

Theorem 4.2 (Exact degeneracy to basic OF)**.** *Assume $k = 1$; the EMPCA basis spans the template direction, $u = \gamma s_0$ with $\gamma \neq 0$; OF and EMPCA use the same weighting Σ^{-1}; and preprocessing and alignment conventions are matched. Then rank-1 EMPCA and basic OF yield identical reconstructed signals $\hat{s}$ and identical weighted residuals.*

Proof sketch. Substituting $u = \gamma s_0$ into the rank-1 coefficient $\hat{c} = (u^\dagger \Sigma^{-1} x)/(u^\dagger \Sigma^{-1} u)$ gives $\hat{c} = \hat{A}_{\mathrm{OF}}/\gamma$, hence $\hat{s}_{\mathrm{EMPCA}} = \hat{c}\,u = \hat{A}_{\mathrm{OF}} s_0 = \hat{s}_{\mathrm{OF}}$. The scalar γ is the rank-1 gauge freedom: it changes the coefficient convention but not the reconstruction. Details in Appendix D.1. □

4.3 Why $k > 1$ helps: the χ^2 improvement

In whitened coordinates, OF projects onto a one-dimensional subspace $\mathcal{S}_1$ and rank-k EMPCA onto a k-dimensional subspace $\mathcal{U}_k$. If $\mathcal{S}_1 \subseteq \mathcal{U}_k$, projection theory gives

$$\chi^2_{\mathrm{EMPCA},k} \leq \chi^2_{\mathrm{OF}} : \tag{21}$$

a larger admissible subspace can only reduce the weighted residual. The physical content is that pulse-shape variation, timing jitter, and channel-coupling modes are additional signal directions that a fixed template conflates into a single amplitude.

Caveat: containment is not automatic. The bound (21) requires the template direction to lie in the learned subspace. Unsupervised EMPCA includes $\mathcal{S}_1$ whenever the signal is the dominant variance direction in whitened space (specifically, when $\mathbb{E}[a^2]\|\tilde{s}\|^2$ exceeds the $(k{+}1)$-st noise eigenvalue); below that SNR, or with unfavorable initialization, containment can fail in finite samples. Template-seeded initialization or an explicit containment constraint restores the guarantee. Section 7 verifies empirically that containment holds in the benchmark regime.

4.4 Geometric interpretation and finite-sample stability

After whitening, both estimators are orthogonal projectors:

$$\hat{\tilde{x}}_{\mathrm{OF}} = P_{\tilde{s}_0}\tilde{x}, \qquad \hat{\tilde{x}}_{\mathrm{EMPCA}} = P_{\tilde{U}}\tilde{x}, \qquad P_{\tilde{U}} = \tilde{U}(\tilde{U}^\dagger\tilde{U})^{-1}\tilde{U}^\dagger. \tag{22}$$

OF projects onto a fixed rank-1 direction; EMPCA onto a learned rank-k subspace; when $k = 1$ and the learned direction equals the template direction, the projectors coincide. Because the objective depends on $\tilde{U}$ only through the projector, EMPCA optimizes over the Grassmannian of k-dimensional subspaces; that geometric picture, and its connection to the latent gauge group, is developed in Appendix E.

In finite samples, the estimated subspace deviates from the population one by an angle controlled by the Davis–Kahan theorem [16]:

$$\sin\theta \leq \frac{\|\hat{R}_x - R_x\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{23}$$

where Δ_λ is the spectral gap between signal and noise eigenvalues of the whitened covariance. For Gaussian noise the numerator is $O(\sqrt{d/N})$ with high probability, and for the rank-1 model the gap grows with SNR as $\Delta_\lambda \approx N_\Phi\,\mathbb{E}[a^2]$. This explains the near-unity weighted cosines observed on the internal calibration sample in Section 7: at high SNR and large N, the sample subspace is extremely close to the population subspace. The extended stability discussion is given in Appendix E.

From algorithm to objective. As defined above, EMPCA is an algorithmic procedure — alternating E- and M-steps. For comparing estimators at the level of the loss, and for replacing the linear encoder by a nonlinear one while preserving the noise-aware structure, an objective-first formulation is preferable. The next section provides it: the same estimator is the minimizer of a noise-aware tied linear autoencoder loss.

5 The Noise-Aware Linear Autoencoder and the Bridge Theorem

5.1 The encoder–decoder language

We now translate the weighted-projection estimators of Sections 3 and 4 into encoder–decoder language. A linear autoencoder consists of an encoder $E : \mathbb{C}^d \to \mathbb{C}^k$ producing a latent code $z = Ex$ and a decoder $D : \mathbb{C}^k \to \mathbb{C}^d$ producing the reconstruction $\hat{x} = Dz$. This is the division of labor already present in OF and EMPCA: OF computes one weighted coefficient along a fixed template and decodes it into a rank-1 waveform; EMPCA computes k weighted coefficients in a learned basis. The autoencoder formulation introduces no new estimator by itself — the question is which loss and which encoder–decoder relationship recover the Gaussian ML projection.

For ordinary Euclidean linear AEs, "tied weights" means the encoder is the Hermitian transpose of the decoder. In detector coordinates this is the wrong adjoint, because the data-space inner product is set by Σ^{-1}, not the Euclidean metric. The noise-aware tied form is

$$D(z) = Pz, \qquad E(x) = P^\dagger \Sigma^{-1} x, \qquad P^\dagger \Sigma^{-1} P = I_k, \tag{24}$$

i.e. the encoder is the Σ^{-1}-adjoint of the decoder. This section shows that this architecture, trained with the Mahalanobis reconstruction loss, is exactly EMPCA.

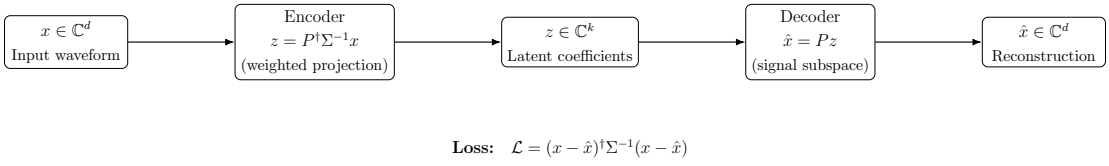


Figure 1: The noise-aware tied linear autoencoder. The encoder is the Σ^{-1}-adjoint of the decoder, and the training loss is the Gaussian χ^2 objective (4).

5.2 The noise-aware loss and its minimizer

With the tied form (24), the reconstruction is $\hat{x}_i = PP^\dagger\Sigma^{-1}x_i$ and the noise-aware reconstruction loss over N events is

$$\mathcal{L}(P) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^\dagger\,\Sigma^{-1}\,(x_i - \hat{x}_i). \tag{25}$$

Expanding with $P^\dagger\Sigma^{-1}P = I_k$,

$$\mathcal{L}(P) = \frac{1}{N}\sum_{i=1}^{N} x_i^\dagger\Sigma^{-1}x_i \;-\; \mathrm{Tr}\big(P^\dagger\Sigma^{-1}\hat{R}_x\Sigma^{-1}P\big), \qquad \hat{R}_x = \frac{1}{N}\sum_i x_i x_i^\dagger, \tag{26}$$

so minimizing $\mathcal{L}(P)$ is a Rayleigh–Ritz problem whose optimum spans the top-k eigenvectors of the Σ^{-1}-weighted covariance operator — precisely the EMPCA subspace. Equivalently, in whitened coordinates ($W = \Sigma^{-1/2}P$, $W^\dagger W = I_k$) the loss is the classical tied linear AE loss $\|\tilde{X} - \tilde{X}WW^\dagger\|_F^2$, minimized by the top-$k$ right singular vectors of $\tilde{X}$.

5.3 The Bridge Theorem

Theorem 5.1 (Bridge Theorem: noise-aware linear AE ≡ EMPCA)**.** *Let $x_i = s_i + n_i \in \mathbb{C}^d$ for $i = 1, \ldots, N$, where $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$ with $\Sigma \succ 0$ known. Consider the noise-aware tied linear autoencoder loss*

$$\mathcal{L}(P) = \frac{1}{N}\sum_{i=1}^{N}(x_i - PP^\dagger\Sigma^{-1}x_i)^\dagger\,\Sigma^{-1}\,(x_i - PP^\dagger\Sigma^{-1}x_i), \qquad P^\dagger\Sigma^{-1}P = I_k. \tag{27}$$

Then:

1. *Every minimizer P^* of $\mathcal{L}(P)$ satisfies* $\mathrm{range}(P^*) = \mathrm{range}(\hat{U})$*, where $\hat{U}$ is the rank-k EMPCA solution with weight Σ^{-1}, up to a unitary rotation within the k-dimensional subspace.*

2. *Equivalently, P^* spans the top-k eigenvectors of the generalized eigenproblem $\hat{R}_x\Sigma^{-1}v = \lambda v$.*

3. *The minimum value is $\mathcal{L}^* = \mathrm{Tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j=1}^{k}\lambda_j$, with $\lambda_1 \geq \cdots \geq \lambda_k$ the top eigenvalues of $\Sigma^{-1}\hat{R}_x$.*

Proof sketch. Whitening maps the constraint to $W^\dagger W = I_k$ and the loss to $\frac{1}{N}\|\tilde{X}(I - WW^\dagger)\|_F^2$. By Rayleigh–Ritz the minimizer is the top-k eigenspace of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, which maps back to the generalized eigenspace of $\hat{R}_x\Sigma^{-1}$; EMPCA converges to the same eigenspace (Section 4). The full proof, including the minimum value and uniqueness up to gauge, is given in Appendix D.3. □

Remark 5.1. *The Bridge Theorem is a finite-sample statement: it holds for any $N \geq 1$, not only in the population limit. The minimizer is unique only up to the latent gauge rotation $P^* \mapsto P^*Q$, $Q \in \mathcal{U}(k)$; physical quantities depend only on the reconstruction $P^*P^{*\dagger}\Sigma^{-1}x$ and are gauge-invariant. Experiment B (Section 7) verifies the theorem numerically: the minimum principal-angle cosine between the EMPCA subspace and the exact AE minimizer exceeds* 0.9999 *for all ranks tested.*

The Bridge Theorem closes the equivalence chain of the paper. The noise-aware tied linear AE is not an unrelated neural architecture: in the linear tied case it is the encoder–decoder *parameterization* of the same weighted low-rank reconstruction problem solved by EMPCA, which in turn degenerates to OF at rank 1. It is therefore the correct linear baseline against which nonlinear autoencoders should be compared.

5.4 Why this architecture: a symmetry summary

There is a structural reason the tied Σ^{-1}-adjoint architecture and the χ^2 loss appear together. Two symmetry groups act on the problem: the *noise isometry group* of linear maps preserving the Σ^{-1} inner product, and the *latent gauge group* $\mathcal{U}(k)$ rotating the basis within the learned subspace. Requiring the quadratic reconstruction loss to be invariant under all noise isometries forces the loss weight to be proportional to Σ^{-1} (the Mahalanobis χ^2 loss, which is also the Fisher information metric of the Gaussian family [17]); requiring the reconstruction to be the metric-orthogonal projection onto the learned subspace forces the encoder to be the Σ^{-1}-adjoint of the decoder. Tied weights are thus not a regularization heuristic but an adjointness condition; untied linear encoders generally define oblique projections and break the equivalence to EMPCA. The formal statement (Theorem E.1) and its proof, together with the Grassmannian geometry of the subspace family, are given in Appendix E.

5.5 Scope of the equivalence claim

The equivalences of this section are rigorous for linear subspace models, Gaussian noise likelihood, matched whitening/weighting and preprocessing conventions, and identical rank k. They do not automatically extend to nonlinear autoencoders or mismatched objectives; what does transfer is the loss geometry, which is the subject of the next section.

6 Loss Geometry: Why MSE Can Be Wrong

The Bridge Theorem has an immediate consequence for machine-learning loss design. Under the measurement model $x_i = s_i + n_i$, $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$, the Gaussian negative log-likelihood of a reconstruction $\hat{x}_i$ is, up to constants,

$$\mathcal{L}_{\Sigma^{-1}} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^\dagger \Sigma^{-1} (x_i - \hat{x}_i), \tag{28}$$

whereas the standard isotropic loss used by ordinary PCA and most autoencoders is

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|_2^2. \tag{29}$$

The two coincide (up to scale) exactly when $\Sigma = \sigma^2 I$. Isotropic MSE is therefore not a neutral default: it is the Gaussian likelihood for white noise, and under colored, heteroskedastic, or correlated detector noise it is a *misspecified* likelihood.

The geometric mechanism. The effect of the misspecification is not a rescaling of the objective; the two losses define different inner products and hence different notions of which directions in waveform space matter. Isotropic MSE weights every time or frequency bin equally, so the learned subspace is driven by raw waveform variance — which, under colored noise, is typically dominated by noise-rich, low-information directions. The Σ^{-1}-weighted loss weights residuals by inverse noise power, so the learned subspace is driven by signal-to-noise structure: noisy directions are down-weighted and informative directions emphasized. The same model class, rank, data, and optimizer can therefore converge to substantially different subspaces under the two losses.

Metric reversal. A concrete failure mode follows. Because each objective is optimal in its own geometry, a method can simultaneously achieve *lower raw MSE* and a *worse weighted residual* (equivalently, worse held-out Gaussian likelihood and worse detector-relevant resolution), while learning a subspace nearly orthogonal — in the whitened metric — to the one required for efficient estimation. We refer to this as *metric reversal*: the ranking of methods reverses depending on whether the evaluation metric respects the noise geometry. Selecting models by raw MSE under colored noise can therefore systematically prefer a reconstruction misaligned with the assumed Gaussian noise model.

Testable predictions. The argument yields two predictions that the experiments of Section 7 test directly, keeping the model class, latent rank, training data, preprocessing, and train/test split fixed while changing only the loss geometry:

- under white noise, isotropic PCA and noise-aware EMPCA should recover the same subspace and agree on all metrics;
- under colored noise, the objectives should diverge: the Σ^{-1}-weighted objective should win on weighted residual, held-out likelihood, and energy resolution, while isotropic PCA may win on raw MSE — the metric it optimizes.

The white-noise case acts as an internal control: if the two objectives coincide when $\Sigma \propto I$ but separate under colored noise, the observed effect is attributable to loss geometry rather than to rank, optimization, or preprocessing differences.

7 Experimental Validation

The experiments test progressively less controlled consequences of the likelihood geometry. Synthetic data test estimator identities and model consequences when the generating covariance is known. Public CRESST pulse traces test the within-channel linear reconstruction hierarchy on held-out detector data. Public GWOSC strain then tests a boundary condition: whether correctly implemented PSD-weighted statistics are calibrated when a simple stationary or time-local Gaussian noise model is applied to real interferometer noise. Because neither release records the true signal, a fourth stage uses public TIDMAD data, whose validation files carry an injected reference channel alongside the detector readout and therefore permit scoring a real-data reconstruction against known truth. The stages are not interchangeable. A synthetic identity does not establish real-data calibration, and reconstruction residuals do not establish trigger sensitivity.

7.1 Evaluation conventions

All reconstruction comparisons use the likelihood-aligned weighted residual $(x-\hat{s})^\dagger\Sigma^{-1}(x-\hat{s})$ defined in Eq. (4). Ordinary PCA is also evaluated by raw MSE to expose disagreement between the Euclidean and covariance-weighted geometries. Subspace agreement is measured by principal angles after applying the same whitening convention. Unless otherwise stated, confidence intervals are percentile bootstrap intervals over the paired unit named in the result. Covariance estimates use calibration data disjoint from evaluation traces. For stationary single-channel data, the covariance is represented by its PSD; the discrete Fourier and normalization conventions are held fixed across methods within each comparison.

The comparisons answer different questions. OF versus rank-6 EMPCA primarily tests signal-model capacity: a fixed template is compared with a learned six-dimensional subspace. Ordinary rank-6 PCA versus rank-6 EMPCA holds the nominal subspace dimen-

sion fixed and is therefore the more direct descriptive contrast for covariance weighting. The tied linear-AE comparison is a subspace implementation check. GWOSC PSD and shared-FIR agreements are also implementation checks, whereas held-out score dispersion is a calibration test. We do not use success on one axis as evidence for another.

7.2 Controlled synthetic validation

The controlled suite contains nine frozen tests. S1–S4 and S6–S9 were run with ten seeds; S5 was subsequently repeated as a confirmatory 30-seed analysis. S1 checks OF normalization against the Cramér–Rao bound; S2 checks recovery of the OF direction by rank-1 EMPCA; S3 compares an independent tied linear-AE implementation with the EMPCA optimum; S4 is a white-noise PCA/EMPCA control; S5 tests colored-noise metric reversal; S6 introduces timing variability and varies rank; S7 varies the amount of covariance-calibration data; S8 checks the whitened residual χ^2; and S9 compares full and diagonal multichannel covariances. Frozen configurations and seed-level outputs are archived with the paper evidence.

All nine regression gates passed (Fig. 2). The OF standard deviation divided by its predicted bound was 0.994 (ten-seed 95% interval $[0.985, 1.002]$). The independent tied linear AE reached the EMPCA objective with mean relative gap 1.95×10^{-16}, and PCA and EMPCA in white noise differed by only 1.48×10^{-6} degrees in maximum principal angle. For S5, each seed generated 512 traces, split at the trace level into equal training and test sets with the same generated data supplied to the two rank-2 estimators. The held-out PCA and EMPCA weighted residuals were 48.418 and 47.878, respectively; the paired PCA-minus-EMPCA difference was 0.540 (30-seed bootstrap 95% interval $[0.523, 0.560]$). Raw MSE preferred the opposite reconstruction (2367.99 for PCA and 5388.75 for EMPCA), while the maximum whitened principal angle was 86.07°. Thus the ordering is metric-dependent under the stated synthetic model; 48.418 is a residual value, not a fold degradation. With timing variability, the mean best rank by clean-signal MSE was 4.6, rather than one. At the largest covariance-calibration size the reconstructed uncertainty was 1.032 times the oracle value; the mean whitened residual per degree of freedom was 0.997; and diagonal rather than full multichannel covariance increased uncertainty by a factor 1.025. These results support the implemented consequences of the stated synthetic model, not real-detector adequacy of any covariance estimate.

7.3 Public CRESST within-channel reconstruction

We used the public CRESST-II/III pulse-shape release [13]. The frozen run at commit `9c9f696ff6bca6e6b9729cdb90e41ee92d576787` used all seven channels in the test release. Each run/channel pair was treated independently and sorted by the release time

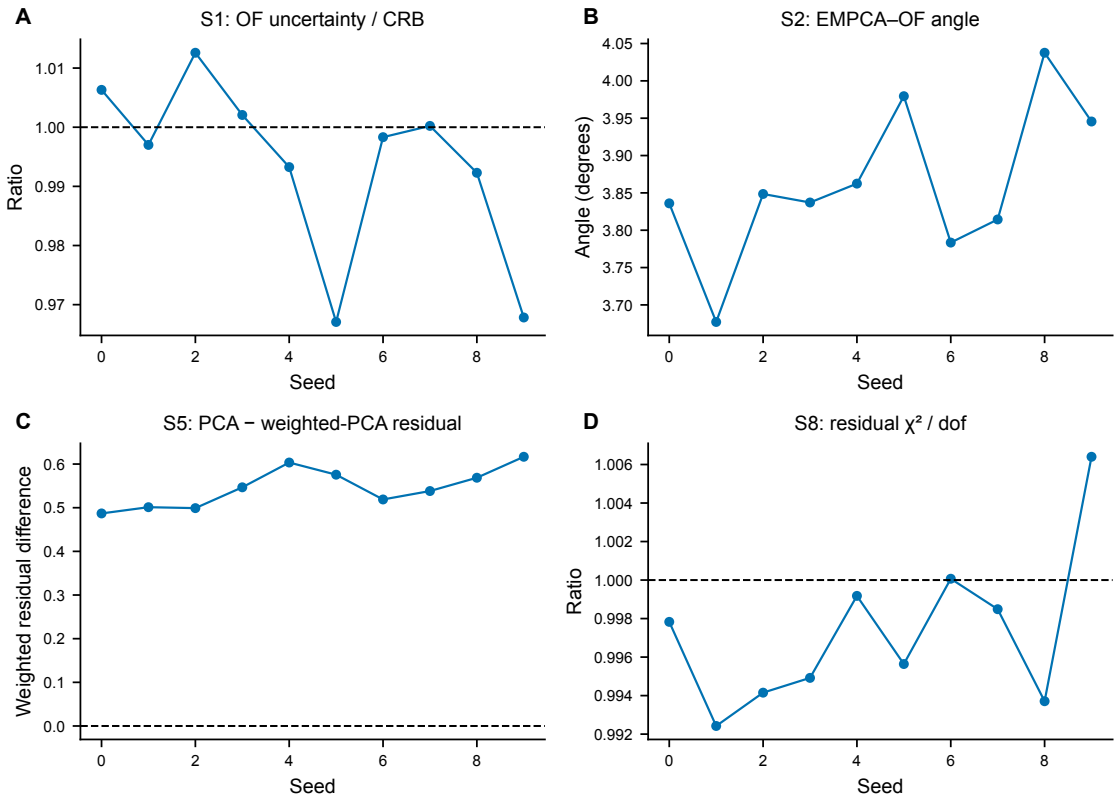


Figure 2: Controlled validation summary. The nine frozen experiments test OF normalization (S1), rank-1 direction recovery (S2), the independent tied-linear-AE/EMPCA bridge (S3), the white-noise control (S4), colored- noise metric reversal (S5), timing-dependent rank (S6), covariance estimation (S7), residual calibration (S8), and multichannel covariance (S9). Panels S1, S2, and S8 show the frozen ten-seed runs; panel S5 shows the confirmatory 30-seed held-out run. Intervals and acceptance rules are archived with the seed-level evidence.

variable. Pulse and noise streams were each split chronologically, separately, using the first 70% for calibration and the remainder for evaluation. Calibration pulses fitted the signal models and calibration noise fitted the PSD; no evaluation pulse entered a fitted object. The separate stream splits do not impose a common cross-stream timestamp cutoff: six and three evaluation pulses in channels `35/57.0` and `35/58.0`, respectively, preceded the last calibration-noise timestamp. Within each stream, row and trace-hash overlap was zero. Depending on channel, calibration contained 3,195–12,692 pulses and 512–537 noise traces, while evaluation contained 1,369–5,440 pulses. The pooled held-out comparison comprised 19,335 pulses.

The frozen signal model was rank-6 EMPCA with a stationary PSD covariance and shrinkage 0.1. Baselines were single-template OF, a multi-template OF bank, ordinary PCA, and an independently optimized tied linear AE. The template bank used rise- and decay-time quantile cells and integer shifts of $-4, -2, 0, 2$, and 4 samples. The primary OF comparison was the paired difference $R_{\mathrm{OF}} - R_{\mathrm{EMPCA}}$. The predeclared competitiveness margin was $1.1R_{\mathrm{multiOF}} - R_{\mathrm{EMPCA}}$. Positive values favor EMPCA. The primary uncertainty calculation pooled paired event-level differences and resampled the 19,335 evaluation pulses 300 times. This event-bootstrap gate is retained as the primary predeclared analysis.

We also performed a post-freeze channel-aware sensitivity analysis from the archived event residuals. Its point estimand gives each of the seven run/channel clusters equal weight. For uncertainty, each of 10,000 bootstrap replicates (seed 20260713) sampled seven clusters with replacement and then sampled the observed number of events with replacement within each selected cluster occurrence. This tests robustness to unequal channel sizes; it does not replace the pooled primary gate.

Rank-6 EMPCA improved the weighted residual relative to single-template OF in every channel (Fig. 3A). The pooled difference was 1012.27 with 95% event-bootstrap CI $[927.90, 1097.61]$. EMPCA also met the multi-template competitiveness gate in every channel: the pooled margin was 352.17 with CI $[272.40, 424.01]$. Because channel residual scales differ, the equal-channel means were 731.87 for OF minus EMPCA and 279.26 for the competitiveness margin. Their channel-then-event 95% intervals were $[110.69, 1537.48]$ and $[36.52, 569.01]$, respectively. Thus both directions survived channel balancing, although seven clusters from two runs make these percentile intervals coarse and potentially unstable.

To separate effect direction from the heterogeneous residual scale, we also formed descriptive ratios from the archived channel means. The fractional reduction $1 - R_{\mathrm{EMPCA}}/R_{\mathrm{OF}}$ ranged from 89.7% to 98.8% across channels, with an equal-channel mean of 96.2%. This large contrast primarily measures the benefit of replacing a fixed template with a rank-6 signal model. In the same-rank geometry contrast,

$1 - R_{\mathrm{EMPCA}}/R_{\mathrm{PCA}}$ was positive in all seven channels but ranged from 1.6% to 62.0%, with an equal-channel mean of 21.6%. The unnormalized equal-channel PCA-minus-EMPCA difference was 3.78, with hierarchical 95% interval $[1.22, 6.69]$. The normalized ratios remain descriptive functions of channel means rather than additional acceptance gates. This heterogeneous result supports a narrower conclusion than the OF comparison: covariance weighting helped the likelihood-aligned residual on these channels, but the magnitude was channel-dependent.

All five negative controls degraded at the pooled level (Fig. 3B). The weakest was wrong-PSD weighting, whose paired difference from EMPCA was 1.57 with CI $[0.97, 2.08]$. The important channel-level exception was run/channel `35/58.0`, for which wrong-PSD weighting did not degrade: the mean difference was approximately -0.005 with CI $[-0.010, 0.0001]$. The other controls had larger pooled mean degradations: 489.40 for noise-only training, 4013.47 for a random weighted basis, 6911.82 for a mismatched OF template, and 59812.72 for permuted time bins. These scale-dependent values are reported to verify direction and falsification behavior, not to compare the scientific severity of different controls. All five control differences also had positive post-freeze hierarchical intervals, despite the single wrong-PSD channel exception.

The independently fitted tied linear AE and EMPCA bases agreed to numerical precision on the public traces: the maximum principal angle across channels was 1.91×10^{-6} degrees (Fig. 3C). This is a real-data consistency check for the tied linear bridge only. Trigger efficiency was inconclusive over the frozen amplitude grid for six of seven channels at the primary 5% false-positive rate, and no channel reached 90% efficiency. We therefore draw no detector-sensitivity conclusion from this stage.

7.4 Public GWOSC calibration boundary

The GWOSC stage used public 4-kHz H1 and L1 strain around GW150914 [14] as a stress test, not an event search. The selected 256-second interval supplied 62 valid off-source windows per detector. A Hann-windowed, bias-corrected median-Welch PSD was estimated from calibration windows and template-score dispersion was evaluated on held-out windows, over five random splits (46 calibration, 16 evaluation) and five chronological blocks (36 calibration, 12–13 evaluation); a null spread divided by its PSD prediction near one was the calibration target. Two implementation checks accompanied it: comparison of the repository PSD estimator with GWpy [18] on identical inputs, and a shared-FIR check forcing explicit FFT convolution and GWpy convolution onto the same FIR coefficients and normalization. A frozen follow-up then compared the global PSD with local radii of 32, 64, and 96 seconds, the 64-second model and acceptance range $[0.8, 1.2]$ being predeclared as primary.

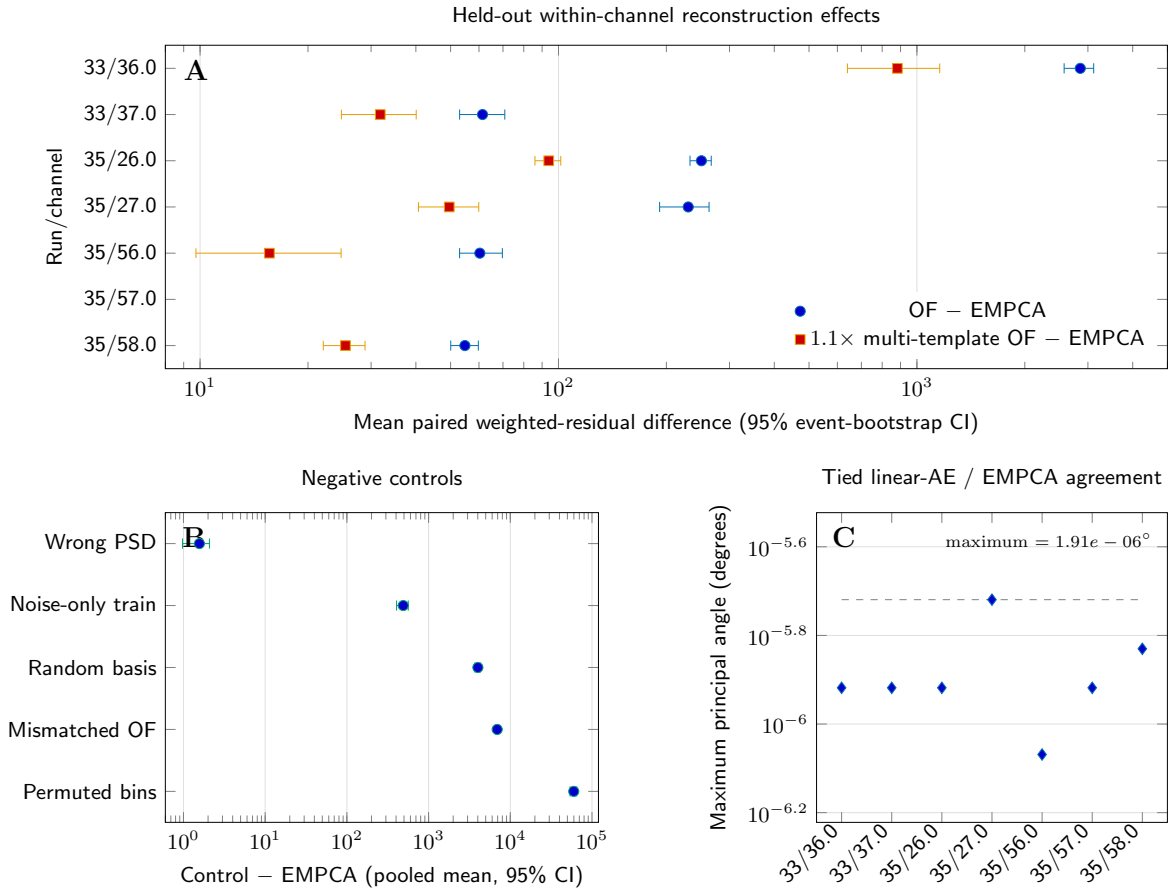


Figure 3: Public CRESST within-channel validation. (A) Per-channel paired effects for single-template OF and the predeclared multi-template-OF competitiveness margin; horizontal bars are 95% event-bootstrap intervals (300 resamples). The logarithmic axis is used because residual scales differ between channels. (B) Pooled paired degradations for five negative controls. Wrong-PSD weighting is the weakest pooled control; its sole sign reversal was on `35/58.0` (mean approximately -0.005, CI $[-0.010, 0.0001]$). (C) Maximum principal angles between independently fitted tied linear-AE and EMPCA subspaces. All values are generated from the archived run JSON; $n = 19{,}335$ held-out pulses across seven channels. Event-bootstrap intervals are conditional on the observed channels; channel-aware sensitivity intervals are reported in the text.

The implementation checks passed and the calibration tests did not (Table 3). PSD ratios were one at recorded precision and the shared-FIR paths agreed far below their 10^{-10} tolerances, yet the global PSD failed the aggregate random and chronological gates for both detectors, with individual null-spread ratios reaching 10.480 for H1 and 7.619 for L1. The predeclared 64-second local model passed its stationary synthetic control but not real H1 or L1, and therefore did not repair calibration. Event scores and nominal injections remain archived diagnostics; we do not convert them to significance, false-alarm probability, or validated sensitivity.

This outcome is a boundary result rather than a verdict on the geometry, and it is the reason for the next stage. Because GWOSC carries no injected reference, a calibration failure there cannot be separated from the question the theory actually asks—whether the covariance-weighted reconstruction is closer to the true signal. Answering that on real data requires a detector release in which the truth is recorded alongside the readout.

Table 3: Compact GWOSC boundary results. PSD agreement and shared-FIR identity are implementation checks. Global and local score-spread results are statistical calibration checks on the selected interval.

Check	Synthetic/reference	H1	L1
PSD ratio, identical windows	reference = 1	1.000	1.000
Shared-FIR max relative L_2 difference	1.47×10^{-15}	1.70×10^{-13}	3.04×10^{-13}
Global null-spread ratio, random median	—	1.986	1.263
Global null-spread ratio, chronological median	—	3.183	2.964
Local-64 score standard deviation	1.148 (pass)	9.424 (fail)	8.827 (fail)

The shared-FIR result has a deliberately narrow interpretation: two software paths agree when they implement the same finite-duration statistic. It does not establish mathematical equivalence between the original PSD-domain GLS statistic and a finite FIR statistic. Likewise, agreement with GWpy for the PSD estimator isolates one implementation question but cannot make a global or local Gaussian PSD model statistically adequate for nonstationary real noise.

7.5 Public TIDMAD ground-truth validation

Neither CRESST nor GWOSC records what the true signal was. Both therefore compare residuals rather than distances to truth, which is the quantity the metric-reversal claim is actually about. The public TIDMAD release from the ABRACADABRA axion search [15] closes that gap: its validation files record an injected reference channel alongside the SQUID readout, at 10 MHz sampling, so a reconstruction can be scored against a real injected signal in real detector noise. We use the release's own channel convention

(channel0001 readout, channel0002 reference). The frozen operating point was 4096-sample windows, rank 2, 512 windows split 70/30 chronologically into 358 fit and 150 evaluation windows with a four-window guard gap, and 300 bootstrap resamples; window-hash and window-index overlap between fit and evaluation were both zero.

As an implementation and data-integrity anchor, the release's own unmodified benchmark script was run on the full 20-file validation set before any analysis. The none baseline reproduced at coarse score 1.1026 against a published 1.10, fixing channel semantics, data integrity, and environment. The savgol baseline did not (-2.7708 against a published -2.35), which is a property of the published hyperparameters rather than of this reproduction: at polyorder 11 the Savitzky–Golay coefficient sum collapses from approximately one to 1.7×10^{-10}, so that baseline is not bit-reproducible by any implementation. We treat the none match as the operative check and make no bit-exact reproduction claim for savgol.

On real science-file noise with a synthetic injection, all six estimator- identity gates passed: fractional amplitude bias -3.0×10^{-4} ($z = -0.68$), empirical over PSD-predicted standard deviation 0.9986, a null consistent with zero ($z = -0.78$), 2.33° maximum principal angle between the learned rank-2 subspace and the analytic quadrature pair, the $\cos^2$ SNR-loss law to within 0.019, and a subspace-error scaling exponent of -0.9996 against the predicted -1. This validates the whitening and estimator machinery on real noise; it is not the metric-reversal test.

Table 4: Compact TIDMAD results. Gate G0 and Phase 1 are implementation checks; Phase 2 is the ground-truth reconstruction comparison; Phase 4 is a calibration check; Phase 5 is the release's own detector-facing score. Values are taken from the archived run JSON at commit 59db37151d45f18e67d2d60142a7140eaf02f47e.

Check	Reference	Observed	Status
G0 none coarse score	published 1.10	1.1026	anchor matches
G0 savgol coarse score	published -2.35	-2.7708	ill-conditioned baseline
Phase 1 six estimator-identity gates	—	all passed	implementation check
Phase 2 truth-error difference, file 0010	> 0	$+0.0036$, CI $[-0.0040, 0.0133]$	direction only
Phase 2 truth-error difference, file 0005	> 0	$+0.0402$, CI $[-0.0793, 0.1779]$	direction only
Phase 4 global-model χ^2/dof	band $[0.9, 1.1]$	1.0031	calibrates
Phase 4 score dispersion	band $[0.9, 1.1]$	1.0142	calibrates
Phase 5 coarse denoising score	no-denoising 1.10	-1.8025	below baseline

Reconstruction against real injected truth. The primary comparison scores rank-2 PCA and rank-2 EMPCA reconstructions of the readout channel against the file's own injected reference.

[PLACEHOLDER — pending evidence: TIDMAD Phase 2 on abra_validation_0000.h5.

Insert the paired truth-error difference and its window- and block-level intervals only after the corresponding run JSON is synchronized into `data/tidmad/current/` and its acceptance status is reviewed.]

Two further validation files are archived, and on both the predicted qualitative pattern held while the paired interval did not exclude zero. On `abra_validation_0010.h5`, raw MSE preferred PCA (171.15 against 199.36), the weighted residual preferred EMPCA (9.147×10^{13} against 9.197×10^{13}), and the injected reference also preferred EMPCA (amplitude RMSE 19.4799 against 19.4836); the paired absolute-error difference was +0.0036, with 95% window- and block-level intervals $[-0.0040, 0.0133]$ and $[-0.0089, 0.0127]$. The same three orderings held on `abra_validation_0005.h5` (raw MSE 131.13 against 148.93; weighted residual 8.105×10^{13} against 8.058×10^{13}; amplitude RMSE 4.8103 against 4.7522), with paired difference +0.0402 and interval $[-0.0793, 0.1779]$. Both are consistent in direction with metric reversal and neither establishes it alone. Their limited resolving power is documented: the calibration tone couples into the reconstructed channel with ratio 3.22 on file `0000` but only 0.057 on file `0010`, the coupling falling off steeply with injection frequency across the release.

Calibration of the noise model. Unlike the GWOSC interval, a simple global Gaussian PSD model calibrates this detector's noise. On `abra_validation_0000.h5` over 512 windows of 4095 degrees of freedom each, the global model gave χ^2 per degree of freedom 1.0031 and held-out score dispersion 1.0142, both inside the predeclared $[0.9, 1.1]$ band and close to the stationary control's own 1.0002 and 1.0140. Local models at radii of 8, 16, and 32 windows gave dispersions of 1.0500, 1.0535, and 1.0276, so no local refinement was needed, and the amplitude cost of the mismatched model was a standard-deviation inflation of 1.0002. This is a positive result about this file and its approximately 0.2-second sample range, not about SQUID readout in general. It does not weaken the GWOSC finding: together the two say that whether a stationary Gaussian model is adequate is an empirical property of the instrument rather than of the framework.

Detector-facing denoising score. Scored through the release's own unmodified benchmark script, the noise-aware linear Wiener estimator achieved a coarse denoising score of -1.8025, below the no-denoising baseline of 1.10 and below every published baseline except the ill-conditioned `savgol` (moving average 0.64; FCNet 6.55; PUNet 3.84; Transformer 4.18 [15]). The predeclared question was how much of the gap between no denoising and a trained deep denoiser a correct noise geometry with a linear model explains, and the honest answer at this operating point is a negative amount. A likelihood-optimal rank-2 estimator for a narrowband injection is not a broadband time-series denoiser, and we make no competitiveness claim on this metric.

Rank prediction and negative controls. The analytic rank-2 prediction, that an unknown-phase sinusoid spans exactly $\mathrm{span}\{\cos, \sin\}$, is confirmed in theory and in controlled synthetic data. Three attempts to confirm it on real TIDMAD validation data did not succeed—including one file selected by a data-quality rule frozen before its outcome was known, on which rank-2 amplitude RMSE 4.752 was statistically indistinguishable from rank-1 4.752, paired CI $[-0.056, 0.098]$. The causes identified are unrelated to the estimator: a persistent two-tone structure in one file's reference channel, a richer manifold than the single-frequency model assumes; weak reference-to-readout coupling in another; and, on the file passing every predeclared quality check, most plausibly insufficient power at 150 evaluation windows. We therefore claim no real-data confirmation of the rank prediction. Of five predeclared negative controls, three degraded the primary metric with 95% CI excluding zero—wrong injection frequency (+0.357), permuted frequency bins (+0.016), and wrong PSD weighting (+0.017), against a nominal amplitude RMSE of 0.0880. Two did not: noise-only training was flat (-0.0025, CI spanning zero), and a random weighted basis improved slightly (-0.0062, CI entirely negative) because it collapses to the zero-recovery level, whose RMSE of 0.0747 is itself below the nominal, leaving that control saturated rather than informative. As with the CRESST wrong-PSD exception, we report both rather than adjusting the controls.

8 Discussion

The theory and experiments support one organizing statement: under the stated linear Gaussian model, Σ^{-1} defines reconstruction geometry. Once that metric is fixed, OF, weighted low-rank reconstruction, and the tied noise-aware linear autoencoder are not competing likelihoods. They are different constraints or parameterizations of the same likelihood projection. OF uses a fixed template, EMPCA learns a subspace, and the tied linear autoencoder writes the same subspace projection as an encoder and decoder. This unification is exact only under matched preprocessing, weighting, normalization, and linear model assumptions.

The resulting loss-design consequence is practical. Isotropic MSE assumes white Gaussian noise. When the covariance is structured, it emphasizes raw variance rather than inverse-noise-weighted information, so its model ranking can disagree with the ranking implied by the assumed likelihood. The controlled white-noise agreement and colored-noise metric reversal isolate this mechanism without changing rank or data. Covariance-weighted residuals are therefore the appropriate primary comparison when a Gaussian covariance model is the intended statistical description. This conclusion does not say that the Gaussian model is always adequate; the GWOSC result shows why model adequacy must itself be tested.

The public CRESST result supplies the positive real-detector component that the synthetic theory alone cannot provide. Every one of the seven channels favored rank-6 EMPCA over a single-template OF on held-out weighted residual, and every channel met the multi-template-OF competitiveness margin. The channel-balanced OF difference was 731.87, with a post-freeze hierarchical 95% interval $[110.69, 1537.48]$. That result mainly tests the signal-model hierarchy: rank-6 EMPCA can represent held-out pulse-shape variation that a fixed template cannot. It should not be attributed wholly to covariance weighting. The same-rank PCA–EMPCA contrast addresses weighting more directly. EMPCA had the lower channel-mean weighted residual in all seven channels, but its fractional advantage ranged from 1.6% to 62.0%. The covariance effect is therefore directionally consistent and heterogeneous, rather than uniformly large; its unnormalized channel-balanced difference was 3.78 with hierarchical 95% interval $[1.22, 6.69]$. The negative controls and independently fitted tied-linear-AE subspaces check the pipeline and linear bridge, respectively; the near-zero AE/EMPCA angle is an implementation-level consistency result, not independent external evidence for the theorem.

The CRESST claim should consequently remain narrow. The basis and PSD were estimated and evaluated within each channel. The experiment supports a within-channel linear reconstruction hierarchy for the released pulses and provides descriptive real-data evidence for covariance weighting. It does not demonstrate transfer to an unseen channel, a physical energy-resolution improvement, or trigger sensitivity. The inconclusive trigger grid reinforces the separation between reconstruction and detection.

The GWOSC analysis establishes a complementary boundary. Agreement with GWpy for identical PSD windows shows that the repository PSD calculation is not trivially different from its reference. Agreement between explicit FFT and GWpy convolutions under shared FIR coefficients and normalization shows software identity for that common finite statistic. Neither check implies calibration. Held-out score spreads were too large under the global PSD, and the fixed 64-second local PSD also failed on real H1 and L1 although its stationary synthetic control passed. Thus implementation correctness is necessary but not sufficient: calibrated likelihood inference additionally requires a noise model that describes the evaluation distribution. The predeclared empirical hypothesis that the 64-second local PSD would restore calibration was falsified on this interval. This contradicts that application- level hypothesis, not the conditional linear-Gaussian projection results. The present experiment does not identify whether the remaining mismatch is due to non-stationarity, transients, data-quality structure, or statistic-specific finite-window effects.

The TIDMAD stage was added because CRESST and GWOSC share a structural limitation: neither records the signal that was actually present, so both compare residuals under an assumed covariance rather than distances to a known truth. TIDMAD's injected reference channel removes that circularity. Its contribution is specific and partial.

The estimator identities hold on real SQUID noise; on the archived validation files the three metrics ordered as predicted, with raw MSE preferring one reconstruction while the weighted residual and the injected truth preferred the other, though the paired intervals on those files include zero and the primary file's result is not yet in the evidence archive. The calibration finding is the clearer one, and it is a positive result rather than a boundary lesson: a simple global Gaussian PSD model calibrates this detector's noise, where the same class of model failed on GWOSC strain. Read together, the two say that stationarity is an empirical property of the instrument and not a defect of the framework, which is a more useful statement than either result alone.

Where TIDMAD did not deliver should be stated with equal directness. The rank-2 quadrature prediction was not confirmed on any of the three real files attempted, and the noise-aware linear estimator scored below the no-denoising baseline on the release's own detector-facing benchmark. Neither outcome contradicts the theory—the diagnoses point to multi-tone structure, weak channel coupling, limited statistical power, and a mismatch between a narrowband rank-2 estimator and a broadband denoising score—but neither supports it either, and we do not present them as if they did.

Together these results clarify the level at which the framework should be used. The covariance-weighted objective is the correct likelihood geometry conditional on Σ. Empirical validation must then ask three separate questions: whether the constrained signal model reconstructs held-out signals better within that geometry, whether the estimated covariance actually calibrates residuals or scores on the target data, and whether the preferred reconstruction is in fact closer to the true signal. CRESST answers the first positively for the within-channel linear hierarchy. GWOSC answers the second negatively for two frozen PSD models on one interval, and TIDMAD answers it positively for one file of a different instrument. The third question is the one only a release with injected truth can address.

9 Limitations

The equivalence results concern linear subspaces and Gaussian noise with a known or independently estimated positive-definite covariance. They do not cover curved signal manifolds, saturation, pile-up, heavy-tailed contamination, or covariance estimation from the same event being reconstructed. Timing variation can be approximated locally by additional linear directions, but a finite subspace need not represent large shifts or other nonlinear waveform changes. Likewise, a Mahalanobis objective is efficient only when its noise model is adequate; covariance drift or outliers can invalidate its uncertainty interpretation even when the optimization is implemented correctly.

Noise covariance also has an identifiability limit that is separate from the linear recon-

struction results. An eigendecomposition of a covariance or cross-spectral-density matrix produces decorrelated statistical modes, not uniquely labelled physical sources. Nothing in this paper recovers a physical noise decomposition from a measured total spectrum, and no result should be read as attributing residual improvement to a named noise mechanism. Where a component budget is supplied with a dataset, its labels are an input to the analysis rather than an output of it, and a conditional attribution computed under such a budget does not establish that the component is separable in hardware or that removing it would improve a real detector by the same amount.

The CRESST inference is based on seven run/channel clusters. The primary predeclared interval resamples 19,335 pooled events and therefore quantifies event-level uncertainty conditional on these channels; it does not quantify the variation that would arise from sampling a new detector channel. We show all channel effects and an equal-channel point estimate to expose scale heterogeneity. A post-freeze channel-then-event analysis with 10,000 resamples retains the primary effect directions, but its intervals are coarse and potentially unstable because only seven clusters from two runs are available. Pulse and noise streams were each split chronologically, separately; the design does not use a common cross-stream timestamp cutoff. It does not test alternative split fractions, cross-channel generalization, or independent acquisition campaigns. One negative-control exception—wrong-PSD weighting on `35/58.0`—also shows that pooled control behavior need not hold for every channel.

The CRESST trigger-efficiency calculation was underpowered over the frozen amplitude grid. Six channels were inconclusive at the primary 5% false-positive rate, and none reached 90% efficiency. Reconstruction gains cannot therefore be translated into a threshold, exposure, or detector- sensitivity improvement. Such a claim would require a separately frozen signal-injection and background protocol with an amplitude range that brackets the relevant efficiencies.

The GWOSC evidence covers one 256-second interval centered on GW150914. Windows and chronological blocks from that interval are not independent astrophysical replications. The global and local PSD failures are interval-specific boundary results, not evidence that the framework fails on all gravitational-wave data. Conversely, the passed PSD and shared-FIR checks do not validate event significance. The original PSD-domain GLS statistic and a finite-duration FIR statistic were not shown to be mathematically equivalent. New background models, data-quality rules, or frequency choices must be frozen and tested on untouched intervals rather than tuned until this interval passes.

The TIDMAD evidence is narrower than the number of phases run suggests. Each result comes from named validation files at a single frozen operating point, and no result should be read as a survey of the release; the injected frequency sweeps within each file and the reference-to-readout coupling varies by more than three orders of magnitude across

files, so file-to-file transfer is not assumed. The calibration finding covers one file and an approximately 0.2-second sample range. The rank-2 prediction has no real-data confirmation: three attempts failed, and although each has an identified non-implementation cause, a diagnosed non-confirmation is not a confirmation deferred. The denoising score is a genuine negative under its predeclared claim boundary. Two of five negative controls did not degrade, one of them because it had saturated at the zero-recovery level, which means the control set here demonstrates falsifiability but cannot rank control severity. The primary Phase 2 comparison remains a working-draft placeholder until its run JSON is synchronized into the evidence archive; the manuscript is not submission-ready while that placeholder stands.

10 Code and data availability

The CRESST pulse-shape data are publicly described in Ref. [13]; the GWOSC event data are available under the GW150914 data-release DOI [14]. The corrective evidence archive uses schema `paper-revision-evidence/1.0.0`, bundle identifier `paper1-revision-20260726-partial-v1`, and manifest SHA-256 `aa7ea1632983d3c59318fca5c1abd6288f0ebd13281d5a97920692e561e52944`. It records experiment-repository commit `34112bfc313a127840a05e17ce93f23720457f14` and the confirmatory S5 run `S5-30seed-heldout-20260726`. All results reported in this paper come from public datasets; no internal or unreleased detector data, noise budget, or instrument simulation is used or described. The CRESST run is tied to source commit `9c9f696ff6bca6e6b9729cdb90e41ee92d576787`, seed 22, and the archived configuration `p2_cresst_validation_seed22.config.yaml`. The post-freeze CRESST sensitivity archive is indexed under `20260713_channel_robustness_v1`; it records 10,000 hierarchical resamples with seed 20260713, lossless event residuals, analysis code, an environment record, a leakage audit, and checksums. The GWOSC baseline archive has run identifier `20260622T175125Z_b169c1f595a4` and base commit `b169c1f595a46d2701417ff4cbce292330817ad2`; the shared-FIR and local-PSD follow-up configurations are archived alongside their JSON outputs. The TIDMAD data are the public release `github.com/jessicafry/TIDMAD` under CC-BY-4.0 [15]; gate G0 used the full twenty-file validation set, and the reported phases used validation files `abra_validation_0000.h5`, `abra_validation_0005.h5`, and `abra_validation_0010.h5` together with the release's science files for real-noise windows. The frozen TIDMAD operating point was 4096-sample windows, rank 2, a 70/30 chronological split with a four-window guard gap, and 300 bootstrap resamples with block size 16, at seed 20260727 and source commit `59db37151d45f18e67d2d60142a7140eaf02f47e`; the channel convention, per-file injection frequencies, and valid sample ranges are recorded

in the archived data contract. The Phase 2 run on `abra_validation_0000.h5` is cited in the manuscript as a pending placeholder and is not yet present in the evidence archive. Raw public strain and pulse arrays are not duplicated in the manuscript archive; source locations and evidence checksums are retained so that the inputs can be retrieved from their public releases and the reported results audited. Because the primary TIDMAD Phase 2 run and the legacy supplementary-experiment provenance remain unresolved, this archive supports a corrective manuscript build but not an arXiv revision-candidate designation.

11 Conclusion

Under a linear Gaussian measurement model, inverse noise covariance defines the geometry of reconstruction. In that geometry, OF is a fixed rank-1 likelihood projection, weighted low-rank reconstruction or EMPCA learns the projection subspace, and the tied noise-aware linear autoencoder recovers the same subspace under the stated conditions. Isotropic MSE is the corresponding likelihood only for white noise. Controlled simulations support these model consequences, while public CRESST data support the narrow claim that rank-6 weighted subspace reconstruction improves within-channel held-out residuals relative to fixed-template baselines; channel-balanced hierarchical sensitivity analyses preserve the effect directions. The smaller and heterogeneous same-rank PCA–EMPCA contrast is the direct real-data evidence about covariance weighting. Public GWOSC data show the complementary limit: correct PSD and shared-statistic implementations were insufficient for calibrated inference under the tested global and local PSD models. Public TIDMAD data, which unlike the other two releases record an injected reference signal, supply the estimator identities on real SQUID noise and a global noise model that calibrates where GWOSC's did not; on that release the rank-2 prediction was not confirmed and the linear estimator did not compete on the detector-facing denoising score. Cross-channel transfer, detector sensitivity, calibrated GW150914 significance, a real-data rank confirmation, and broader real-noise generalization remain outside the evidence reported here.

References

[1] George L. Turin. An introduction to matched filters. *IRE Transactions on Information Theory*, 6(3):311–329, 1960.

[2] Steven M. Kay. *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Prentice Hall, 1993.

[3] A. E. Szymkowiak, R. L. Kelley, S. H. Moseley, and C. K. Stahle. Signal processing for microcalorimeters. *Journal of Low Temperature Physics*, 93(3–4):281–285, 1993.

[4] Dale J. Fixsen, Samuel H. Moseley, Blas Cabrera, and Enectali Figueroa-Feliciano. Pulse estimation in nonlinear detectors with nonstationary noise. *Nuclear Instruments and Methods in Physics Research A*, 520:555–558, 2004. Optimal filter derivation for calorimetric detectors.

[5] Noah Kurinsky et al. Optimal filter derivation and implementation for cryogenic detectors. Technical report, CDMS/SuperCDMS Collaboration, 2021. Internal technical note, Appendix E.

[6] Michael E. Tipping and Christopher M. Bishop. Probabilistic principal component analysis. *Journal of the Royal Statistical Society: Series B*, 61(3):611–622, 1999.

[7] Donald B. Rubin and Dorothy T. Thayer. EM algorithms for ML factor analysis. *Psychometrika*, 47(1):69–76, 1982.

[8] Stephen Bailey. Principal component analysis with noisy and/or missing data. *Publications of the Astronomical Society of the Pacific*, 124(919):1015–1023, 2012.

[9] Ningxin Yu et al. Optimal-filter pulse analysis for microcalorimeters with nonstationary noise. *Journal of Low Temperature Physics*, 199:718–726, 2020.

[10] Pierre Baldi and Kurt Hornik. Neural networks and principal component analysis: Learning from examples without local minima. *Neural Networks*, 2(1):53–58, 1989.

[11] Hervé Bourlard and Yves Kamp. Auto-association by multilayer perceptrons and singular value decomposition. *Biological Cybernetics*, 59:291–294, 1988.

[12] Elad Plaut. From principal subspaces to principal components with linear autoencoders. *arXiv preprint arXiv:1804.10253*, 2018.

[13] CRESST Collaboration, G. Angloher, S. Banik, D. Bartolot, G. Benato, et al. Description of CRESST-II and CRESST-III pulse shape data. *arXiv preprint arXiv:2508.03078*, 2025.

[14] LIGO Scientific Collaboration and Virgo Collaboration. Data release for event GW150914. Gravitational Wave Open Science Center, 2016. Accessed 12 July 2026.

[15] J. T. Fry, J. L. Ouellet, C. P. Salemi, L. Winslow, et al. TIDMAD: Time series dataset for discovering dark matter with AI denoising, 2024. Public release `github.com/jessicafry/TIDMAD`, CC-BY-4.0.

[16] Chandler Davis and William M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.

[17] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, Providence, RI, 2000.

[18] Duncan M. Macleod, Joseph S. Areeda, Scott B. Coughlin, Thomas J. Massinger, and Alexander L. Urban. GWpy: A Python package for gravitational-wave astrophysics. *SoftwareX*, 13:100657, 2021.

[19] Gene H. Golub, Michael Heath, and Grace Wahba. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.

[20] Paul Tseng. Convergence of a block coordinate descent method for nondifferentiable minimization. *Journal of Optimization Theory and Applications*, 109(3):475–494, 2001.

[21] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478v2*, 2021.

[22] Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2):303–353, 1998.

[23] P.-A. Absil, Robert Mahony, and Rodolphe Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.

A Detailed Optimal-Filter Derivations and the Cramér–Rao Bound

This appendix expands the summary statements of Section 3.2: unbiasedness, variance, Fisher information, and the energy-resolution formula for the OF estimator.

Unbiasedness. Taking the expectation of $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ under the model $X_k = AS_k + N_k$:

$$\mathbb{E}[\hat{A}] = \sum_k \Phi_k^{(0)} A S_k = A \sum_k \frac{S_k^* S_k}{J_k N_\Phi} = A \cdot \frac{N_\Phi}{N_\Phi} = A. \tag{30}$$

Estimator variance. Because the noise bins N_k are independent with $\mathbb{E}[|N_k|^2] = J_k$,

$$\mathrm{Var}(\hat{A}) = \sum_k |\Phi_k^{(0)}|^2 J_k = \sum_k \frac{|S_k|^2}{J_k^2 N_\Phi^2} J_k = \frac{1}{N_\Phi^2} \sum_k \frac{|S_k|^2}{J_k} = \frac{1}{N_\Phi}, \tag{31}$$

which is Eq. (14) of the main text.

Fisher information and the CRB. The Fisher information for the scalar parameter A from a single trace is

$$\mathcal{I}(A) = -\mathbb{E}\left[\frac{\partial^2 \log \mathcal{L}}{\partial A^2}\right] = 2 \sum_k \frac{|S_k|^2}{J_k} = 2N_\Phi, \tag{32}$$

where the factor of 2 arises from the real–imaginary decomposition under the complex Gaussian model (and is absent in a real-valued model; conventions can also differ by a factor of 2 between one-sided and two-sided PSDs — see Appendix G). The CRB is $\mathrm{Var}(\hat{A}) \geq 1/\mathcal{I}(A)$, and equality holds for $\hat{A}$, confirming that the OF estimator is efficient and hence the minimum-variance unbiased estimator in the Gaussian model.

Energy resolution. With amplitude calibrated to energy, $A = E/E_0$ for a known calibration energy E_0 (e.g. a K-α X-ray), the baseline energy resolution is

$$\sigma_E = E_0 \cdot \sigma_A = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}, \tag{33}$$

which is Eq. (15). Resolution improves when signal power is concentrated at frequencies where the noise PSD is small; N_Φ is both the filter normalization and the total signal-to-noise budget. The noise-budget study of Appendix F uses this formula to rank noise components by their contribution to σ_E.

Bias under template mismatch. When the assumed template s_0 differs from the true signal direction s,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0}, \tag{34}$$

so the estimator is biased toward zero in proportion to the whitened-space mismatch. The variance formula (31) is then no longer the CRB for the true signal; the effective SNR retention is

$$\frac{(s_0^\dagger \Sigma^{-1} s)^2}{(s_0^\dagger \Sigma^{-1} s_0)(s^\dagger \Sigma^{-1} s)} = \cos^2 \theta_w, \tag{35}$$

the squared cosine of the whitened angle between s_0 and s. In whitened coordinates, the OF projector captures a fraction $N_\Phi / \mathrm{Tr}(\Sigma^{-1} \hat{R}_x)$ of the total signal-to-noise budget; when the true family is not one-dimensional, the remaining SNR is distributed across additional directions that rank-k EMPCA recovers.

Reference implementation. For completeness, the single-event OF amplitude estimator is:

1. Compute DFTs $X_k \leftarrow \mathrm{FFT}(x)$, $S_k \leftarrow \mathrm{FFT}(s_0)$.
2. Numerator $\mathcal{N} \leftarrow \mathrm{Re}\big(\sum_k S_k^* X_k / J_k\big) \cdot (2/N f_s)$.
3. Denominator $\mathcal{D} \leftarrow \sum_k |S_k|^2 / J_k \cdot (2/N f_s)$.
4. $\hat{A} \leftarrow \mathcal{N}/\mathcal{D}$; residual $\chi^2 \leftarrow \sum_k |X_k - \hat{A} S_k|^2 / J_k \cdot (2/N f_s)$.

All operations are $O(d)$ after the one-time $O(d \log d)$ FFT. The factor $2/(N f_s)$ converts DFT-normalized quantities to physical one-sided PSD units (Appendix G). For time-shift OF, step 1 is augmented by a grid search over candidate shifts with $S_k \to S_k e^{-2\pi i k \tau_j / N}$, selecting the shift that minimizes χ^2 (Appendix B.1).

B Time-Shift and Joint-Channel Optimal Filtering

This appendix gives the full derivations for the two OF extensions summarized in Section 3.3, and proves the degeneracy propositions referenced there.

B.1 Time-shift optimal filtering

Include an unknown arrival-time shift t_0 via

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \qquad X_k = A\, S_k(t_0) + N_k, \tag{36}$$

with weighted objective

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - Ae^{-i\omega_k t_0} S_k|^2}{J_k}. \tag{37}$$

For fixed t_0, the stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A}(t_0) = \frac{\sum_k e^{i\omega_k t_0} S_k^* X_k / J_k}{\sum_k |S_k|^2 / J_k} = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k, \tag{38}$$

so $\hat{A}(t_0)$ is the inverse DFT of the filtered spectrum [5]. To profile out the amplitude, define

$$\chi_0^2 = \sum_k \frac{|X_k|^2}{J_k}, \qquad B(t_0) = \sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}. \tag{39}$$

Then

$$\chi^2(A, t_0) = \chi_0^2 - A^* B(t_0) - A B^*(t_0) + |A|^2 N_\Phi, \tag{40}$$

and at $A = \hat{A}(t_0) = B(t_0)/N_\Phi$,

$$\chi^2(t_0) = \chi_0^2 - \frac{|B(t_0)|^2}{N_\Phi} = \chi_0^2 - |\hat{A}(t_0)|^2 N_\Phi. \tag{41}$$

Hence the optimal delay is

$$\hat{t}_0 = \arg\min_{t_0} \chi^2(t_0) = \arg\max_{t_0} |\hat{A}(t_0)|, \tag{42}$$

the usual "maximize filtered amplitude in time" criterion for real-valued traces with Hermitian spectra.

Relation to learned subspaces. The shifted templates $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$ form a one-parameter signal manifold, and time-shift OF performs ML estimation over this manifold. For small shifts, the Taylor expansion

$$e^{-i\omega t_0} s_0 = s_0 - i\omega t_0\, s_0 - \tfrac{1}{2}(\omega t_0)^2 s_0 + \cdots \tag{43}$$

shows that the shift family is approximately contained in $\mathrm{span}\{s_0, \omega s_0, \omega^2 s_0, \dots\}$ — in time-domain notation, $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$.

Proposition B.1 (Degeneracy to OF with shift)**.** *OF with time shift is the ML estimator obtained by restricting the signal model to the nonlinear shift manifold $\mathcal{M}$. EMPCA degenerates to shifted OF when the learned subspace is exactly the span of the shift family; more generally, low-rank EMPCA provides a linear-subspace approximation to shifted OF when the dominant variation in the data is timing jitter around a common template.*

Basic OF, shifted OF, and rank-k EMPCA thus form a nested hierarchy: basic OF $\subset$ shifted OF $\subset$ low-rank EMPCA approximation to the shift variation. The former numerical illustration has been omitted from this revision because its seed-level provenance was not recoverable from the archived evidence; the analytical statement above does not depend on it.

B.2 Joint-channel correlated optimal filtering

Joint observation model. For C channels with d samples per channel, stack the channel traces:

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \qquad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \tag{44}$$

with noise $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma_{\text{joint}})$ and block-structured covariance

$$\Sigma_{\text{joint}} = \begin{bmatrix} \Sigma^{(11)} & \Sigma^{(12)} & \dots & \Sigma^{(1C)} \\ \Sigma^{(21)} & \Sigma^{(22)} & \dots & \Sigma^{(2C)} \\ \vdots & & \ddots & \vdots \\ \Sigma^{(C1)} & \dots & \dots & \Sigma^{(CC)} \end{bmatrix}, \tag{45}$$

where $\Sigma^{(cc)}$ is the intra-channel covariance and $\Sigma^{(cc')}$ the cross-channel correlation; for stationary noise, $\Sigma^{(cc)}$ is circulant and $\Sigma^{(cc')}$ is the cross-spectral density matrix.

Estimator and degeneracy. The Gaussian ML objective is unchanged in form, $\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma_{\text{joint}}^{-1}(x - \hat{s})$. The rank-1 constraint $\hat{s} = As$ gives the joint-channel correlated optimal filter

$$\hat{A}_{\text{JOF}} = \frac{s^\dagger \Sigma_{\text{joint}}^{-1} x}{s^\dagger \Sigma_{\text{joint}}^{-1} s}, \tag{46}$$

which is algebraically the rank-1 weighted subspace estimator (7) with $U = s$.

Proposition B.2 (Degeneracy to joint-channel correlated OF)**.** *Weighted EMPCA with the full inverse covariance $\Sigma_{\text{joint}}^{-1}$ is the multi-channel subspace generalization of joint-channel correlated optimal filtering. In the rank-1 case, when the EMPCA basis spans the joint template direction, weighted EMPCA reduces exactly to joint-channel correlated OF.*

EMPCA does not change the underlying likelihood; it replaces a fixed joint template by a learned joint signal subspace, with noise correlations handled identically through the quadratic form.

Whitening and decoupling. The joint whitening transform is $\tilde{x} = \Sigma_{\text{joint}}^{-1/2} x$, computed by Cholesky or eigendecomposition. For stationary noise with per-frequency cross-spectral matrices $\Sigma_f \in \mathbb{C}^{C\times C}$, whitening decomposes into independent $C \times C$ Cholesky factors per frequency bin, reducing to C scalar divisions when channels are uncorrelated. If $\Sigma^{(cc')} = 0$ for $c \neq c'$ and each $\Sigma^{(cc)}$ is diagonal, the joint objective decouples into C independent single-channel problems: joint-channel OF becomes the Fisher-information-weighted mean of per-channel OF estimates, and multi-channel EMPCA becomes independent single-channel EMPCA on the whitened channels.

C EMPCA Updates, Convergence, and Conditioning

This appendix collects the algorithmic content supporting Section 4: the EM updates and their derivation, the frequency-domain variant, the coordinate-descent identity with the AE objective, the convergence proof, and the practical machinery (initialization, model-order selection, whitening numerics, and conditioning).

C.1 Classical PCA and its noise assumption

For a mean-subtracted data matrix, classical PCA seeks a rank-k model $X \approx C^\dagger U^\dagger$ (rows of X are events) minimizing the Euclidean reconstruction error $\|X - C^\dagger U^\dagger\|_F^2$ subject to $U^\dagger U = I_k$; the solution is the span of the top-k eigenvectors of the sample covariance. PCA is the ML subspace estimator only under isotropic noise $\Sigma = \sigma^2 I$; any colored or heteroskedastic noise breaks this optimality. EMPCA [8] restores ML optimality by carrying the weight Σ^{-1} (or per-observation weights) through the estimation.

C.2 EM updates

Write the joint objective in the variables (U, C),

$$\mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} (x_i - U c_i)^\dagger \Sigma^{-1} (x_i - U c_i), \tag{47}$$

which is also the tied-AE loss of Section 5 with the encoder constraint imposed at convergence.

E-step (coefficients). For fixed U, $\mathcal{L}$ is strictly convex and quadratic in each c_i; the unique minimizer solves the weighted normal equations

$$(U^\dagger \Sigma^{-1} U)\, c_i = U^\dagger \Sigma^{-1} x_i, \qquad \hat{c}_i = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x_i, \tag{48}$$

which reduces to $\hat{c}_i = U^\dagger \Sigma^{-1} x_i$ under the Σ^{-1}-orthonormality constraint — exactly the tied-AE encoder.

M-step (basis). For fixed C, with aggregates $G = CC^\dagger/N$ and $H = CX\Sigma^{-1}/N$, the unconstrained minimizer is

$$U_{\text{unc}} = H^\dagger G^{-1}, \tag{49}$$

followed by re-orthonormalization in the Σ^{-1} inner product (thin QR of $\Sigma^{-1/2} U_{\text{unc}}$, then unwhitening) — a generalized Gram–Schmidt step that does not increase the objective.

Coordinate-descent view. Each step is the exact block minimizer of (47), so

$$\mathcal{L}(U^{(r+1)}, C^{(r+1)}) \le \mathcal{L}(U^{(r)}, C^{(r+1)}) \le \mathcal{L}(U^{(r)}, C^{(r)}), \tag{50}$$

i.e. the EMPCA E- and M-steps are coordinate descent on the noise-aware AE loss. This is the algorithmic face of the Bridge Theorem.

Deflation variant. Equivalent to Bailey's one-component deflation, components may be solved sequentially, updating $X \leftarrow X - u_m c_m$ before solving the next component.

Reference iteration. Rank-k EMPCA with weighting Σ^{-1}:

1. Initialize $U^{(0)}$ (random orthonormal, SVD warm-start, or template-seeded; see below).
2. E-step: solve (48) for all events ($O(k^2 dN)$).
3. M-step: form (49); re-orthonormalize via QR of $\Sigma^{-1/2} U$ ($O(kdN) + O(k^2 d)$).
4. Evaluate $\mathcal{L}^{(r+1)}$; stop when the relative decrease is below tolerance.

For $k = 1$ the iteration reduces to a weighted power method and converges in $O(1)$ iterations from a template seed.

C.3 Frequency-domain variant and dual interpretation

Yu et al.'s microcalorimeter setting [9] motivates a frequency-domain variant robust to random time shifts. In Fourier space, $X_k^{(j)} = A_j e^{-i\omega_k t_{0,j}} S_k + N_k^{(j)}$, and direct PCA is confounded because the effective coefficient depends on the frequency bin. Using phase differences $\Delta\phi_k = \phi_k - \phi_{k-1}$, a time shift contributes $(\omega_k - \omega_{k-1}) t_0$, which is constant for uniform frequency spacing; the transformed coefficients then behave as a single complex factor per event, restoring a linear subspace model.

With per-element weights W_{kj} (e.g. inverse PSD), the weighted EMPCA objective is

$$\chi^2_{\mathrm{EMPCA}} = \sum_k \sum_j W_{kj} \left| X_{kj} - [UC]_{kj} \right|^2 , \tag{51}$$

with weighted E-step normal equations $(U^\dagger W_j U) c_j = U^\dagger W_j X_{:,j}$ per event and weighted M-step row solves $(C W_k C^\dagger) u_k = C W_k X^\dagger_{k,:}$ per frequency. For stationary noise the covariance is diagonal in the Fourier basis, Σ^{-1} acts element-wise, and weighted inner products reduce to Hadamard products: the Hadamard-product implementation is the exact dual PCA formulation under a diagonal noise covariance, not an approximation.

C.4 Amplitude estimation details

Gauge-invariant amplitude estimators. Because the basis is identified only up to $\mathcal{U}(k)$ rotation, physical amplitudes use gauge-invariant functionals: the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}} = \|c\|_2$ (under $U^\dagger \Sigma^{-1} U = I_k$), calibrated against a reference line; or the projection onto a calibration template, Eq. (20).

Ridge-regularized coefficient solves. When k exceeds the effective rank of the data, or basis vectors are nearly collinear in whitened space, $U^\dagger \Sigma^{-1} U$ becomes ill-conditioned. Tikhonov regularization

$$\hat{c}_\lambda = (U^\dagger \Sigma^{-1} U + \lambda I_k)^{-1} U^\dagger \Sigma^{-1} x \tag{52}$$

controls the bias–variance tradeoff; λ may be chosen by generalized cross-validation [19], though for $N \gg k$ a small numerical floor suffices after proper re-orthonormalization.

C.5 Convergence proof

Theorem C.1 (Monotone convergence of EMPCA)**.** *Let $\mathcal{L}^{(r)}$ denote the value of the noise-aware objective* (18) *after iteration r of the EMPCA algorithm. Then (i) $\mathcal{L}^{(r+1)} \le \mathcal{L}^{(r)}$ for all r; (ii) the sequence $\{(U^{(r)}, C^{(r)})\}$ has at least one limit point; and (iii) every limit point is a fixed point of the alternating-minimization map, i.e. a stationary point of* (18)*.*

Proof. The E-step solution (48) is the unique global minimizer of a strictly convex quadratic, so $\mathcal{L}(U^{(r)}, C^{(r+1)}) \le \mathcal{L}(U^{(r)}, C^{(r)})$. The M-step minimizes the quadratic over U; re-orthonormalization cannot increase the objective because the constrained minimizer is no worse than any feasible point. Chaining gives monotone decrease (50); since $\mathcal{L} \ge 0$, the sequence converges. The iterates live in a compact set (orthonormality constraint plus bounded coefficients), so a limit point exists. The update map $F : (U, C) \mapsto (U^{(r+1)}, C^{(r+1)})$ is continuous; along a subsequence converging to a limit

point (U^*, C^*), $\mathcal{L}(F(U^*, C^*)) = \lim \mathcal{L}^{(r+1)} = \mathcal{L}(U^*, C^*)$, so $F(U^*, C^*) = (U^*, C^*)$ up to gauge, and every fixed point of F satisfies the first-order stationarity conditions of (18). □

Remark C.1 (Global vs. local optima)**.** *The objective is non-convex jointly in (U, C) for $k \geq 2$, but all stationary points correspond to the same k-dimensional eigenspace of $\Sigma^{-1}\hat{R}_x$ (absent eigenvalue degeneracies), so the reconstructed signal is the same at every stationary point; in practice EMPCA converges to it from almost any initialization.*

Remark C.2 (Rate)**.** *When $\lambda_{k+1} < \lambda_k$, alternating minimization converges locally at a linear rate with asymptotic factor $\rho \leq \lambda_{k+1}/\lambda_k$, by the quadratic structure of the objective and standard block-coordinate-descent theory [20]. The empirical convergence traces in Appendix F are consistent with this spectral-gap rate.*

C.6 Initialization

Random: draw $U^{(0)}$ Haar-uniformly on the Stiefel manifold; simple, effective when $N \gg kd$; multiple restarts confirm convergence to the same subspace. *SVD warm-start:* top-k right singular vectors of the unweighted data matrix; a good warm start when noise is not far from isotropic, and empirically 3–5× faster than random (Appendix F). *Template-seeded:* set $u_1 \propto \tilde{s}/\|\tilde{s}\|$ from a calibration template, extending to rank k with random whitened-orthogonal directions; guarantees the containment condition of Section 4.3 and converges essentially immediately for $k = 1$.

C.7 Model-order selection

Three complementary criteria estimate the intrinsic signal dimension k^*. *Scree gap:* in whitened coordinates the noise floor has expected eigenvalue 1, so signal eigenvalues are visibly elevated; $k^* = \arg\max_j \lambda_j/\lambda_{j+1}$. *Information criteria:*

$$\mathrm{AIC}(k) = -2\log\hat{\mathcal{L}}_k + 2p_k, \qquad \mathrm{BIC}(k) = -2\log\hat{\mathcal{L}}_k + p_k \log N, \tag{53}$$

with $p_k = 2kd - k^2$ effective real parameters (the k^2 gauge parameters removed); BIC selects smaller k and is preferred at moderate N. *Cross-validated χ^2:* hold out events and select the k at which out-of-sample χ^2 stops decreasing — the most reliable criterion when N is large. A physical upper bound on k (the number of independent pulse-shape variation modes) usefully narrows the search.

C.8 Whitening numerics and conditioning

Computing $\Sigma^{-1/2}$. For circulant Σ (stationary noise), whitening is element-wise division by $\sqrt{J_k}$ in the frequency domain. Otherwise: eigendecomposition $\Sigma = V\Lambda V^\dagger$ gives

$\Sigma^{-1/2} = V\Lambda^{-1/2}V^\dagger$ (numerically stable, $O(d^3)$); Cholesky $\Sigma = LL^\dagger$ gives $\Sigma^{-1/2} = L^{-\dagger}$ up to a square-root convention (faster, preferred for well-conditioned dense Σ).

Ill-conditioned noise PSDs. Bins with $J_k \approx 0$ amplify noise after whitening. Remedies: (i) an additive floor $J_k \to \max(J_k, \delta \max_k J_k)$ with $\delta \in [10^{-4}, 10^{-2}]$; (ii) frequency masking to a band $[f_{\min}, f_{\max}]$ where the signal has appreciable power — lossless when the signal has negligible power outside the band, and standard in cryogenic detector OF; (iii) Tikhonov regularization $\Sigma^{-1} \to (\Sigma + \varepsilon I)^{-1}$ when the noise floor itself is uncertain.

Ill-conditioned basis matrices. A persistently large condition number of $U^\dagger\Sigma^{-1}U$ despite re-orthonormalization signals k above the effective signal dimension: re-orthonormalize every M-step (standard), ridge the coefficient solve (52), or reduce k.

Computational notes. Sliding-window inference uses fixed eigenvectors with fast coefficient solves, producing amplitude, χ^2, and likelihood traces efficiently. The E-step at $O(k^2dN)$ dominates at large k but is negligible relative to FFT cost at the saturation ranks ($k \leq 4$) relevant to the benchmark detector; batched GPU linear algebra (e.g. batched GEMM and fused weighting/projection kernels) accelerates batch workloads.

D Proofs of the Theorem Statements

D.1 Extended proof of Theorem 4.1

We restate and prove the rank-1 OF–EMPCA equivalence in full detail. We work in whitened coordinates $\tilde{x}_i = \Sigma^{-1/2}x_i$, $\tilde{s} = \Sigma^{-1/2}s$, $\tilde{n}_i = \Sigma^{-1/2}n_i$, so that

$$\tilde{x}_i = a_i\tilde{s} + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}(0, I). \tag{54}$$

Step 1: population covariance is a spiked identity. Because $\mathbb{E}[\tilde{n}_i\tilde{n}_i^\dagger] = I$ and $\mathbb{E}[a_i\tilde{n}_i^\dagger] = 0$,

$$R := \mathbb{E}[\tilde{x}_i\tilde{x}_i^\dagger] = \nu\,\tilde{s}\tilde{s}^\dagger + I, \qquad \nu := \mathbb{E}[a_i^2] > 0. \tag{55}$$

The eigenvalues of R are $\lambda_1 = 1 + \nu\|\tilde{s}\|^2$ (simple, eigenvector $\tilde{s}/\|\tilde{s}\|$) and $\lambda_j = 1$ for $j \geq 2$.

Step 2: EMPCA recovers the leading eigenvector. Rank-1 EMPCA maximizes explained population variance, i.e. finds the leading eigenvector of R:

$$u_1 = \frac{\tilde{s}}{\|\tilde{s}\|}. \tag{56}$$

The eigenvalue gap $\lambda_1 - \lambda_2 = \nu\|\tilde{s}\|^2 > 0$ guarantees uniqueness whenever $\nu > 0$ and $s \neq 0$.

Step 3: the EMPCA projection coefficient. The raw projection coefficient is $u_1^\dagger \tilde{x}_i = \tilde{s}^\dagger \tilde{x}_i / \|\tilde{s}\|$. Rescaling so that a unit-amplitude signal gives $\hat{a}_i = 1$,

$$\hat{a}_i^{(\mathrm{EMPCA})} = \frac{\tilde{s}^\dagger \tilde{x}_i}{\tilde{s}^\dagger \tilde{s}} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}. \tag{57}$$

Step 4: comparison to the OF estimator. Minimizing $\chi^2(a) = (x_i - as)^\dagger \Sigma^{-1} (x_i - as)$ over a gives

$$\hat{a}_i^{(\mathrm{OF})} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}, \tag{58}$$

identical to (57). Subspace equivalence therefore implies amplitude equivalence, which implies estimator equivalence.

Step 5: efficiency. Both estimators are unbiased with common variance

$$\mathrm{Var}(\hat{a}_i) = \frac{1}{s^\dagger \Sigma^{-1} s}, \tag{59}$$

equal to the CRB for model (54), so the shared estimator is efficient. □

D.2 Proof of Theorem 4.2

For rank-1 EMPCA with basis u,

$$\hat{c} = \frac{u^\dagger \Sigma^{-1} x}{u^\dagger \Sigma^{-1} u}. \tag{60}$$

Substituting $u = \gamma s_0$,

$$\hat{c} = \frac{\gamma^* s_0^\dagger \Sigma^{-1} x}{|\gamma|^2 s_0^\dagger \Sigma^{-1} s_0} = \frac{1}{\gamma} \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0} = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}, \tag{61}$$

hence

$$\hat{s}_{\mathrm{EMPCA}} = \hat{c}\, u = \frac{\hat{A}_{\mathrm{OF}}}{\gamma} (\gamma s_0) = \hat{A}_{\mathrm{OF}}\, s_0 = \hat{s}_{\mathrm{OF}}, \tag{62}$$

and the weighted residual $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$ is identical for both estimators. The remaining freedom $u \mapsto \alpha u$, $c \mapsto c/\alpha$ is the rank-1 gauge transformation: it changes the coefficient convention but not the reconstructed signal. □

D.3 Extended proof of the Bridge Theorem (Theorem 5.1)

Step A: reduction to whitened form. Set $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $W = \Sigma^{-1/2} P$; the constraint $P^\dagger \Sigma^{-1} P = I_k$ becomes $W^\dagger W = I_k$, and since $P P^\dagger \Sigma^{-1} x_i = \Sigma^{1/2} W W^\dagger \tilde{x}_i$, the loss (27) becomes

$$\mathcal{L}(W) = \frac{1}{N} \|\tilde{X}(I - W W^\dagger)\|_F^2. \tag{63}$$

Step B: expansion and Rayleigh–Ritz. Expanding,

$$\mathcal{L}(W) = \frac{1}{N}\|\tilde{X}\|_F^2 - \mathrm{Tr}\big(W^\dagger \hat{R}_{\tilde{x}} W\big), \qquad \hat{R}_{\tilde{x}} = \frac{1}{N}\tilde{X}^\dagger \tilde{X} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}, \tag{64}$$

so minimizing $\mathcal{L}$ over $W^\dagger W = I_k$ is the Rayleigh–Ritz problem maximizing $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W)$, solved by the top-$k$ eigenvectors $W^* = [v_1, \ldots, v_k]$ of $\hat{R}_{\tilde{x}}$.

Step C: translation back and connection to EMPCA. With $P^* = \Sigma^{1/2}W^*$: left-multiplying $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}v_j = \lambda_j v_j$ by $\Sigma^{1/2}$ gives $\hat{R}_x\Sigma^{-1}(\Sigma^{1/2}v_j) = \lambda_j(\Sigma^{1/2}v_j)$, the generalized eigenproblem solved by EMPCA at convergence (Appendix C). Hence $\mathrm{range}(P^*) = \mathrm{range}(\hat{U})$ up to a unitary rotation within the subspace.

Step D: minimum value. At the optimum, $\mathrm{Tr}(W^{*\dagger}\hat{R}_{\tilde{x}}W^*) = \sum_{j=1}^k \lambda_j$, so

$$\mathcal{L}^* = \frac{1}{N}\|\tilde{X}\|_F^2 - \sum_{j=1}^k \lambda_j = \mathrm{Tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j=1}^k \lambda_j, \tag{65}$$

the unexplained variance in whitened coordinates. □

Remark on uniqueness. W^* is unique only up to right multiplication by $Q \in \mathcal{U}(k)$. Physical estimators (reconstructed signals, likelihoods, χ^2 values) depend only on the projector $WW^\dagger$, which is invariant under such rotations. The population version of the theorem replaces $\hat{R}_x$ by $\mathbb{E}[xx^\dagger]$ and connects to the asymptotic regime of Theorem 4.1.

E Symmetry, Gauge Freedom, and Geometric Interpretation

This appendix develops the structural background summarized in Section 5.4: the symmetry-axiomatic derivation of the noise-aware architecture, the Grassmannian geometry of the subspace family, and the finite-sample stability statement.

E.1 Symmetry-axiomatic derivation of the noise-aware architecture

Definition E.1 (Noise isometry group)**.**

$$G_{\text{noise}} = \{Q \in GL(\mathbb{C}^d) : Q^\dagger\Sigma^{-1}Q = \Sigma^{-1}\} \tag{66}$$

is the group of linear isometries of the noise inner product $\langle u, v\rangle_{\Sigma^{-1}} = u^\dagger\Sigma^{-1}v$. In whitened coordinates, every $Q \in G_{\text{noise}}$ becomes a unitary map $\tilde{Q} = \Sigma^{-1/2}Q\Sigma^{1/2} \in \mathcal{U}(d)$.

Definition E.2 (Latent gauge group)**.** *For a rank-k decoder basis $P \in \mathbb{C}^{d\times k}$, the latent gauge group is $G_{\text{gauge}} = \mathcal{U}(k)$ acting by*

$$P \mapsto PQ, \qquad z \mapsto Q^\dagger z, \qquad Q \in \mathcal{U}(k), \tag{67}$$

which changes the coordinate basis of the learned subspace but leaves the reconstruction Pz invariant. The physical object is the rank-k signal subspace and its projector, not a particular basis. This is a global *gauge symmetry — one unitary acting on the whole latent space — not a local gauge field over a base manifold in the geometric-deep-learning sense [21].*

Theorem E.1 (Symmetry-axiomatic uniqueness)**.** *Consider linear rank-k reconstructions $\hat{x} = DEx$ with $D(z) = Pz$, $P^\dagger\Sigma^{-1}P = I_k$, and quadratic residual loss $\mathcal{L}_M(x, \hat{x}) = (x - \hat{x})^\dagger M(x - \hat{x})$, $M \succ 0$. Suppose:*

1. noise-metric invariance*: $\mathcal{L}_M(Qx, Q\hat{x}) = \mathcal{L}_M(x, \hat{x})$ for all $Q \in G_{\text{noise}}$;*
2. gauge invariance*: $P \mapsto PQ$, $z \mapsto Q^\dagger z$ leaves $\hat{x}$ and the loss unchanged for all $Q \in \mathcal{U}(k)$;*
3. subspace unbiasedness*: $DEp = p$ for every $p \in \text{range}(P)$;*
4. metric-orthogonal projection*: among rank-k linear maps with range* range(P)*, DE minimizes the noise-metric residual norm.*

Then $M = \lambda\Sigma^{-1}$ for some $\lambda > 0$ (the loss is the Gaussian χ^2 up to scale); the encoder is the Σ^{-1}-adjoint of the decoder,

$$E = D^\sharp = P^\dagger\Sigma^{-1}, \qquad DE = PP^\dagger\Sigma^{-1}; \tag{68}$$

and minimizing the resulting loss over all such rank-k subspaces yields the rank-k EMPCA subspace.

Proof. **Loss uniqueness.** Noise-metric invariance gives $Q^\dagger MQ = M$ for all $Q \in G_{\text{noise}}$. In whitened coordinates, with $\tilde{M} = \Sigma^{1/2}M\Sigma^{1/2}$ and $\tilde{Q} = \Sigma^{-1/2}Q\Sigma^{1/2}$, this reads $\tilde{Q}^\dagger\tilde{M}\tilde{Q} = \tilde{M}$ for every $\tilde{Q} \in \mathcal{U}(d)$. The only matrices commuting with all of $\mathcal{U}(d)$ are scalar multiples of the identity, so $\tilde{M} = \lambda I$ with $\lambda > 0$, i.e. $M = \lambda\Sigma^{-1}$. Equivalently, information geometry selects the same loss: for the Gaussian family with fixed covariance Σ, the Fisher information metric on the mean parameter is Σ^{-1}, and the corresponding squared geodesic distance is the Mahalanobis χ^2 distance [17].

Adjointness and gauge invariance. For $D(z) = Pz$, the adjoint with respect to $\langle\cdot,\cdot\rangle_{\Sigma^{-1}}$ in data space and the Euclidean inner product in latent space is the unique

$D^\sharp$ with $\langle Dz, x\rangle_{\Sigma^{-1}} = \langle z, D^\sharp x\rangle_2$; since $\langle Pz, x\rangle_{\Sigma^{-1}} = z^\dagger P^\dagger \Sigma^{-1} x$, we get $D^\sharp x = P^\dagger \Sigma^{-1} x$. The metric-orthogonal rank-k projection onto range(P) is then $DD^\sharp = PP^\dagger \Sigma^{-1}$ using $P^\dagger \Sigma^{-1} P = I_k$. Under $P \mapsto PQ$ the encoder transforms as $Q^\dagger P^\dagger \Sigma^{-1} x$ and the reconstruction $PQQ^\dagger P^\dagger \Sigma^{-1} x = PP^\dagger \Sigma^{-1} x$ is unchanged, proving both the tied form and the gauge invariance.

EMPCA minimizer. Substituting $E = P^\dagger \Sigma^{-1}$ and $M = \lambda \Sigma^{-1}$ gives the loss (25) up to scale; expanding gives Eq. (26), and minimizing over $P^\dagger \Sigma^{-1} P = I_k$ is the Rayleigh–Ritz problem for the top-k whitened eigenspace — the rank-k EMPCA subspace. □

Corollary E.2 (Bridge Theorem from symmetry)**.** *The noise-aware tied linear AE of Eq.* (27) *is the unique linear rank-k architecture–loss pair satisfying the requirements of Theorem E.1, up to latent gauge rotations and an overall positive scaling of the loss; its minimizer therefore coincides with the rank-k EMPCA subspace, recovering Theorem 5.1.*

Remark E.1 (Tied weights as the adjoint condition)**.** *The tied-weight constraint is not a regularization heuristic; it states that encoder and decoder are mutual adjoints under the correct inner products. Untied linear encoders generally define oblique rather than Σ^{-1}-orthogonal projections and so break the geometric condition that makes the AE objective equivalent to EMPCA.*

Remark E.2 (Connection to geometric deep learning)**.** *In geometric-deep-learning terms [21], G_{noise} is the symmetry group of the observation-space metric induced by the noise model, and $G_{\text{gauge}} = \mathcal{U}(k)$ is the global gauge freedom of the latent basis. The blueprint usually derives architectures from symmetries of the input domain; here the symmetry is information-geometric, imposed by the detector covariance. The Bridge Theorem is thus not an algebraic coincidence but the linear reconstruction method selected by the noise metric and the latent gauge freedom.*

E.2 Grassmannian geometry of the subspace family

When $\tilde{U}^\dagger \tilde{U} = I_k$, the EMPCA objective depends on $\tilde{U}$ only through the orthogonal projector $\Pi_{\tilde{U}} = \tilde{U}\tilde{U}^\dagger$: EMPCA optimizes not over bases but over the space of k-dimensional subspaces of $\mathbb{C}^d$, the complex Grassmannian

$$\mathrm{Gr}(k, d) = \mathcal{U}(d)/(\mathcal{U}(k) \times \mathcal{U}(d-k)), \tag{69}$$

equivalently the space of rank-k orthogonal projectors in whitened coordinates. The $\mathcal{U}(k)$ gauge freedom is precisely the freedom to choose a basis within a fixed point of $\mathrm{Gr}(k, d)$. For $k = 1$, $\mathrm{Gr}(1, d) = \mathbb{C}P^{d-1}$ and basic OF corresponds to the fixed point represented by the whitened template direction $[\tilde{s}_0]$. This places EMPCA within the standard geometry of Grassmannian subspace optimization [22, 23].

E.3 Finite-sample stability: extended Davis–Kahan discussion

The Davis–Kahan bound (23) can be read as a deformation-stability statement for the Grassmannian estimator. Let $f(R)$ map a covariance operator to its leading rank-k eigenspace, a point of $\mathrm{Gr}(k, d)$. Finite sampling perturbs the population covariance,

$$\tau : \; R_x \mapsto \hat{R}_x = R_x + E, \qquad \|E\|_{\mathrm{op}} = O(\sqrt{d/N}), \tag{70}$$

and Davis–Kahan gives

$$\mathrm{dist}_{\mathrm{Gr}}\big(f(\hat{R}_x), f(R_x)\big) \lesssim \frac{\|E\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{71}$$

with the inverse spectral gap as stability constant. Structurally this is the same role played by deformation-stability bounds in geometric deep learning [21]: small perturbations of the estimated object produce small changes in the geometric estimator when the stability constant is controlled. The relevant perturbation here is a *signal-space* deformation — the covariance estimate changes on the fixed observation domain — as opposed to a *domain* deformation as in graph or manifold settings. The numerical agreement between independently fitted tied linear-AE and EMPCA subspaces on the public CRESST channels (Fig. 3) is consistent with this stability statement, although it is an implementation-level subspace check rather than a direct estimate of the spectral gap.

F Supplementary Evidence Status

Earlier drafts included quantitative panels for principal-component interpretation, Cramér–Rao checks, resolution scaling, template mismatch, time-shift recovery, rank saturation, timing-bias reduction, residual calibration, and iterative convergence. The current evidence bundle does not map those displayed values to exact configurations, seeds, machine-readable metrics, and a code revision. In particular, it does not independently substantiate the repeated residual values 1029.2, 1028.9, and 1028.6, the reported scaling slope, the stated timing-bias reduction, or the claimed Kolmogorov–Smirnov transition. Those quantitative claims and their legacy figures are therefore omitted from this revision rather than treated as verified results. The controlled S1–S9 records reported in Section 7 remain the synthetic evidence for the claims made in the main text.

A conditional noise-budget attribution analysis, which would have compared the discrete optimal-filter information computed from a baseline total PSD against the information obtained after removing or rescaling individual additive components, was also prepared for an earlier draft. It is not reported here. Such quantities are conditional attributions under a declared additive model: they do not show that a component can be physically removed, that its label is uniquely identifiable from a total spectrum, or

that hardware performance would improve by the corresponding amount. Reporting one would additionally require the sampling frequency, record length, template normalization, endpoint factors, frequency support, and acquisition transfer function to be frozen and independently verified, which the available preparation record does not establish. The claim is therefore withdrawn from this revision rather than reported conditionally.

G Noise Sampling and Spectral Conventions

This appendix fixes the discrete spectral conventions used by the synthetic experiments of Section 7 and referenced by the optimal-filter derivations of Appendix A. It concerns sampling and normalization only; no detector, instrument, or measured noise budget is described here.

G.1 Generating colored noise from a one-sided PSD

The synthetic noise used throughout is a real Gaussian time-domain trace drawn from a specified one-sided PSD density $P(f)$. For trace length N, sampling frequency f_s, spacing $\Delta f = f_s/N$, and rFFT bins $f_k = k\Delta f$, we adopt the NumPy convention of an unnormalised forward transform with a factor $1/N$ in `irfft`. For each interior positive-frequency bin we draw

$$z_k = \frac{a_k + ib_k}{\sqrt{2}}, \qquad a_k, b_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad \tilde{\eta}_k = \sqrt{\frac{f_s N P_k}{2}}\, z_k, \tag{72}$$

where $P_k = P(f_k)$, so that $\mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k/2$. The DC coefficient and, for even N, the Nyquist coefficient have no independent imaginary part and are drawn as real Gaussians,

$$\tilde{\eta}_k = \sqrt{f_s N P_k}\, a_k, \qquad \mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k. \tag{73}$$

For odd N, the highest positive-frequency bin remains an interior complex bin. With this convention, Parseval's identity gives the discrete one-sided relation

$$\mathbb{E}\left[\frac{1}{N}\sum_{n=0}^{N-1}\eta_n^2\right] = \Delta f \sum_{k=0}^{\lfloor N/2\rfloor} P_k, \tag{74}$$

which is the identity used to convert between DFT-normalized quantities and physical one-sided PSD units in the optimal-filter expressions. Conventions in the literature differ by a factor of two between one-sided and two-sided PSDs, and by a further factor of two between real-valued and complex Gaussian models; the two-factor combination is held fixed across all methods within each comparison so that no ordering reported in this

paper depends on it. The stochastic DC component is set to zero and any deterministic mean is represented separately. Built-in spectral shapes and custom tabular PSDs use the same density-to-bin-power conversion.

G.2 Sampling contract and its verification

Drawing independent coefficients on a finite DFT grid defines a periodic stationary process whose covariance is circulant, not Toeplitz. When the requested covariance is a genuinely non-periodic Toeplitz matrix, a separate exact finite-Toeplitz sampler is used instead; these are distinct sampling contracts and are not interchangeable at finite N. The distinction matters for the residual-calibration test S8, where the whitened residual is compared against its nominal χ^2 distribution.

Figure 4 is a second-order spectral check only: agreement of the ensemble mean spectrum with the target density verifies Eqs. (72)–(74) but does not by itself establish Gaussianity. Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance agreement are tested separately in the generator validation suite and are not reproduced here.

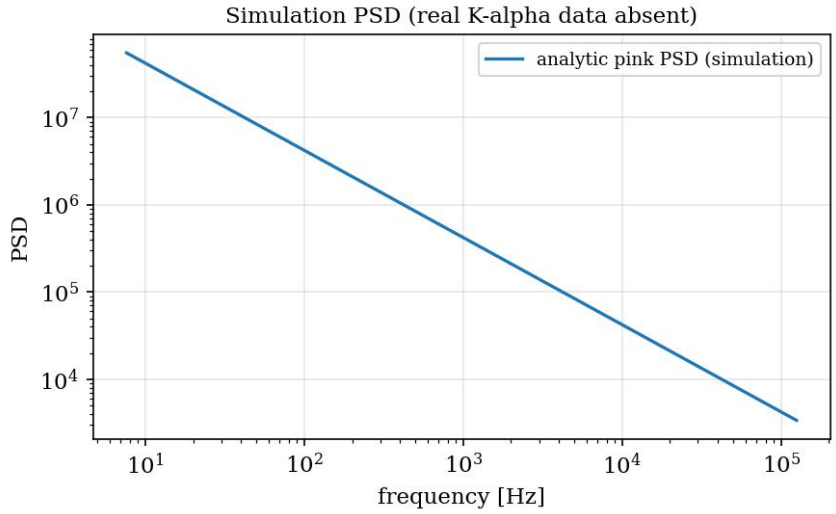


Figure 4: Second-order PSD audit for the one-sided convention of Eq. (72) used by the synthetic OF and EMPCA experiments. Agreement of the ensemble mean spectrum with the target does not by itself establish Gaussianity; Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance are tested separately.

H List of Symbols

Notation used throughout the paper. Complex notation with † (conjugate transpose) is used by default; all statements specialize to real data with $\dagger \to T$. Vectors are columns; whitened quantities carry tildes.

Category	Symbol	Meaning
Dimensions	d	Trace length (samples or frequency bins per event).
	N	Number of events in the dataset.
	k	Rank of the learned signal subspace (also frequency-bin index; clear from context).
	C	Number of detector channels.
Indices	i	Event index, $i = 1, \ldots, N$.
	k	Frequency-bin index of the DFT.
	c	Channel index, $c = 1, \ldots, C$.
	r	EM iteration index.
Data	$x,\ x_i$	Observed trace (column vector in $\mathbb{C}^d$).
	X_k	Frequency-domain coefficient of the observed trace at bin k.
	X	Data matrix, rows $x_i^\dagger$ ($N \times d$).
	$\tilde{x},\ \tilde{X}$	Whitened trace / data matrix, $\tilde{x} = \Sigma^{-1/2} x$.
Signal model	$s_0,\ S_k$	Fixed signal template (time / frequency domain).
	$\hat{s}$	Reconstructed signal.
	$A,\ a_i$	Signal amplitude (pulse height) of an event.
	t_0	Signal arrival time (time offset).
	U	Learned rank-k basis, $U \in \mathbb{C}^{d \times k}$, $U^\dagger \Sigma^{-1} U = I_k$.
	$c,\ C$	Latent coefficient vector / matrix for the rank-k model.
	P	Decoder basis of the noise-aware linear AE, $P^\dagger \Sigma^{-1} P = I_k$.
	z	Latent code of the autoencoder, $z = P^\dagger \Sigma^{-1} x$.

Category	Symbol	Meaning
Noise	$n,\ N_k$	Additive noise (time / frequency domain).
	Σ	Noise covariance matrix ($\succ 0$); diagonal in frequency space for stationary noise.
	Σ^{-1}	Inverse covariance; the observation-space metric.
	$\Sigma^{-1/2}$	Whitening transform.
	$J(f),\ J_k$	Noise power spectral density (PSD).
Optimal filtering	χ^2	Mahalanobis / weighted least-squares objective, Eq. (4).
	$\Phi_k,\ \Phi_k^{(0)}$	Unnormalized / normalized OF kernels.
	N_Φ	Filter normalization $\sum_k \lvert S_k\rvert^2/J_k$; total SNR budget.
	$\sigma_A,\ \sigma_E$	Amplitude / energy resolution.
	$B(t_0)$	Delayed matched-filter response (time-shift OF).
Subspace estimation	$\hat{R}_x$	Sample covariance $N^{-1}\sum_i x_i x_i^\dagger$.
	λ_j	Eigenvalues of $\Sigma^{-1}\hat{R}_x$ (descending).
	Δ_λ	Spectral gap between signal and noise eigenvalues.
	$\rho_w,\ \theta_w$	Weighted (whitened-space) subspace cosine / angle.
	$P_{\tilde{U}}$	Orthogonal projector onto the whitened subspace.
	$\mathrm{Gr}(k,d)$	Grassmannian of k-dimensional subspaces.
Symmetry	G_{noise}	Noise isometry group preserving Σ^{-1} (Definition E.1).
	$\mathcal{U}(k)$	Latent gauge group of basis rotations (Definition E.2).